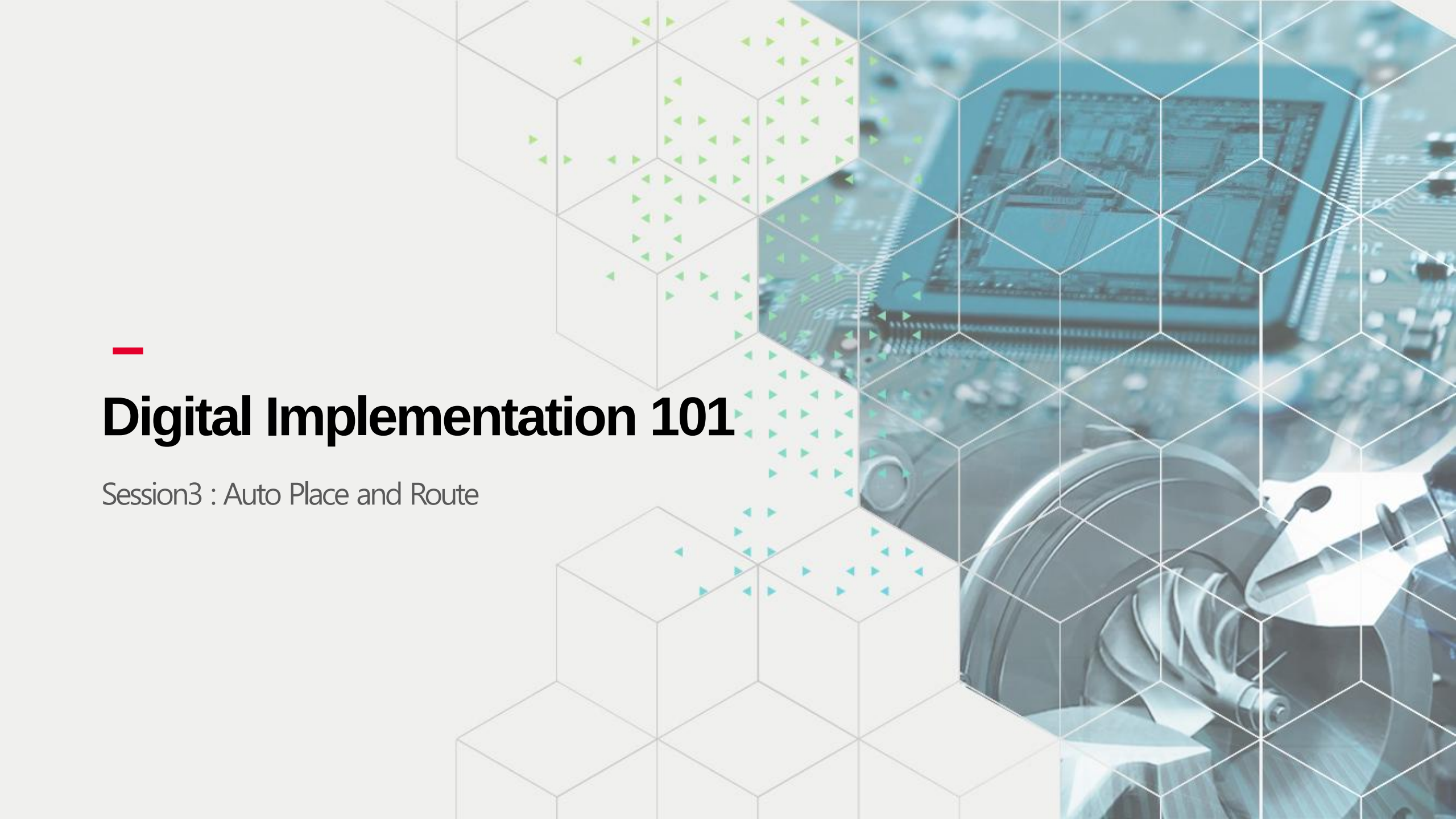


The background features a light gray hexagonal grid pattern. Overlaid on this are two main images: a blue-tinted photograph of a printed circuit board (PCB) with various electronic components, and a close-up of a precision mechanical lathe or mill tool cutting a metal part. Small green and blue triangles are scattered across the grid, particularly concentrated around the circuit board and mechanical images.

Digital Implementation 101

Session3 : Auto Place and Route

1. Auto Place and Route(APR) Flow

Notion of Implementation

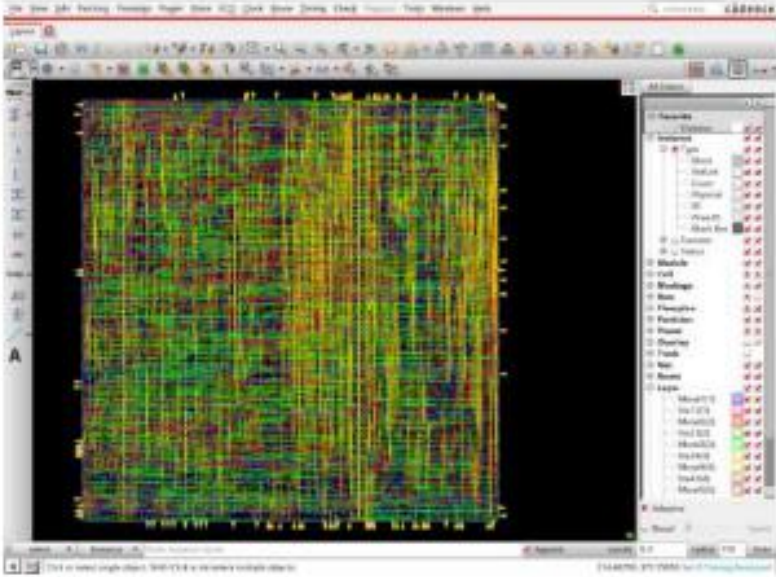
Gate-Level
Netlist



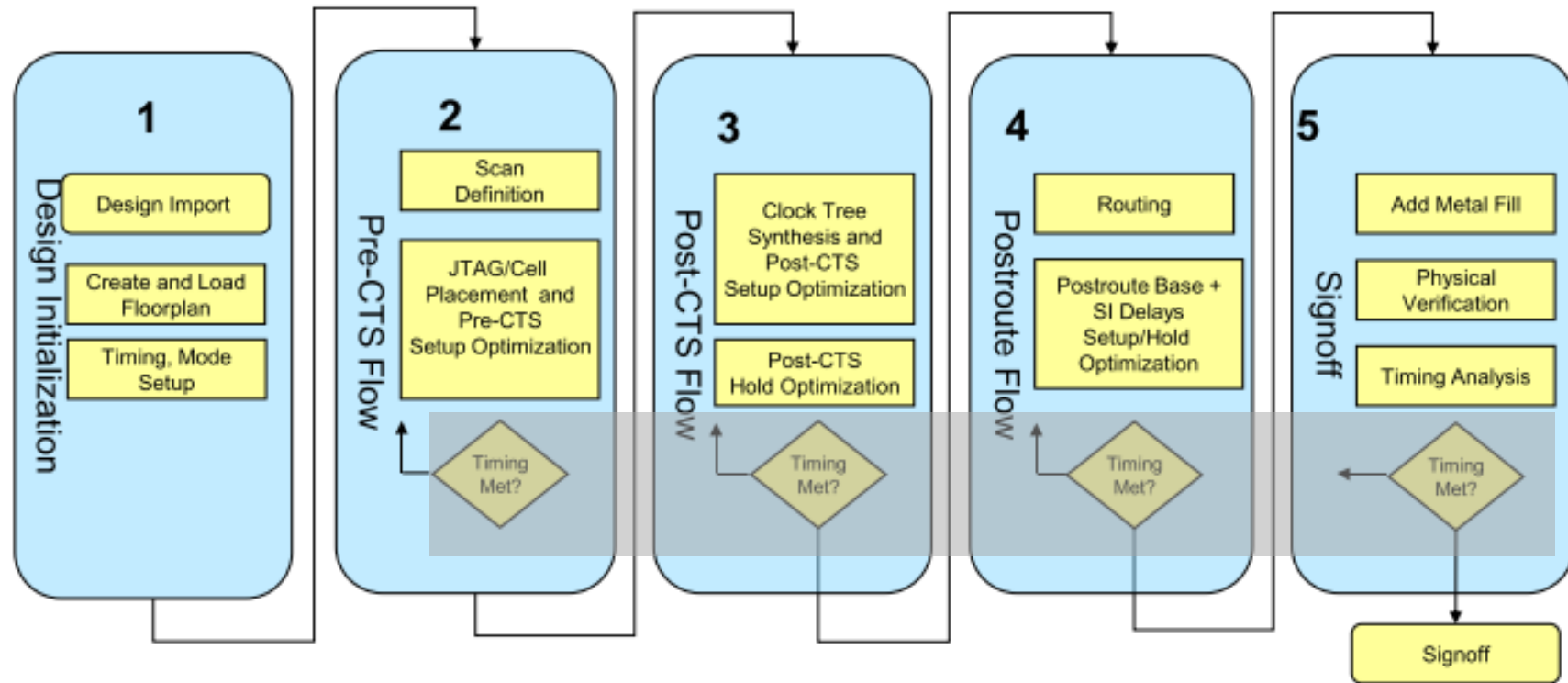
Innovus™
Implementation
System



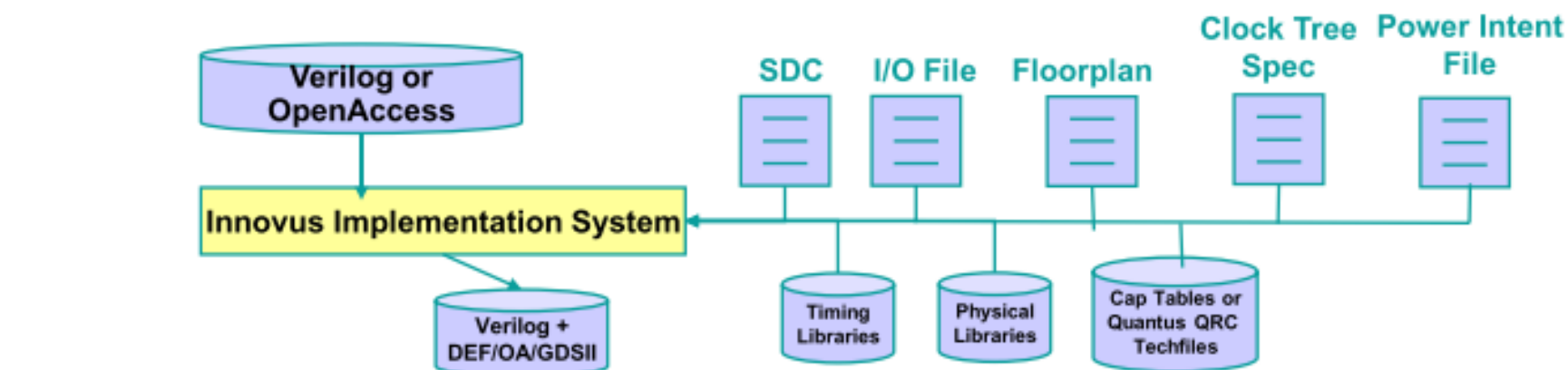
```
module accu(in, acc, clk, reset);
  input [7:0] in;
  input clk, reset;
  output [7:0] acc;
  wire [7:0] in;
  wire clk, reset;
  wire [7:0] acc;
  wire n_0, n_1, n_2, n_3, n_4, n_5, n_6, n_7;
  wire n_8, n_9, n_10, n_11, n_12, n_13, n_14, n_15;
  wire n_16, n_17, n_18, n_19, n_20, n_21, n_22, n_23;
  wire n_24, n_25, n_26, n_27, n_28, n_29, n_30, n_31;
  DFFPOSK1 \acc_reg[0] (.CLK (clk), .D (n_5), .Q (acc[0]));
  DFFPOSK1 \acc_reg[1] (.CLK (clk), .D (n_9), .Q (acc[1]));
  DFFPOSK1 \acc_reg[2] (.CLK (clk), .D (n_13), .Q (acc[2]));
  DFFPOSK1 \acc_reg[3] (.CLK (clk), .D (n_17), .Q (acc[3]));
  DFFPOSK1 \acc_reg[4] (.CLK (clk), .D (n_21), .Q (acc[4]));
  DFFPOSK1 \acc_reg[5] (.CLK (clk), .D (n_25), .Q (acc[5]));
  DFFPOSK1 \acc_reg[6] (.CLK (clk), .D (n_29), .Q (acc[6]));
  DFFPOSK1 \acc_reg[7] (.CLK (clk), .D (n_31), .Q (acc[7]));
  NOR2K1 g146(A (n_30), B (reset), Y (n_31));
  XOR2K1 g148(A (n_26), B (n_3), Y (n_30));
  NOR2K1 g149(A (n_28), B (reset), Y (n_29));
  INVX1 g150(A (n_27), Y (n_18));
  FAX1 g151(A (acc[0]), B (in[0]), C (n_22), .YC (n_26), .YS (n_27));
  NOR2K1 g153(A (n_24), B (reset), Y (n_25));
  INVX1 g154(A (n_23), Y (n_14));
  FAX1 g155(A (acc[1]), B (in[1]), C (n_18), .YC (n_22), .YS (n_23));
  NOR2K1 g157(A (n_20), B (reset), Y (n_21));
  INVX1 g158(A (n_19), Y (n_10));
  FAX1 g159(A (acc[4]), B (in[4]), C (n_14), .YC (n_18), .YS (n_19));
  NOR2K1 g161(A (n_16), B (reset), Y (n_17));
  INVX1 g162(A (n_15), Y (n_16));
  FAX1 g163(A (acc[3]), B (in[3]), C (n_10), .YC (n_14), .YS (n_15));
  NOR2K1 g165(A (n_12), B (reset), Y (n_13));
  INVX1 g166(A (n_11), Y (n_12));
  FAX1 g167(A (acc[2]), B (in[2]), C (n_6), .YC (n_10), .YS (n_11));
  NOR2K1 g169(A (n_8), B (reset), Y (n_9));
  INVX1 g170(A (n_7), Y (n_8));
  FAX1 g171(A (acc[1]), B (in[1]), C (n_1), .YC (n_6), .YS (n_7));
  NOR2K1 g173(A (n_4), B (reset), Y (n_5));
  INVX1 g174(A (n_2), Y (n_4));
  FAX1 g175(A (in[0]), B (acc[0]), .YC (n_1), .YS (n_2));
  OR21K1 g176(A (in[7]), B (acc[7]), C (n_0), Y (n_3));
  NAND2K1 g177(A (acc[7]), B (in[7]), Y (n_0));
endmodule
```



Implementation Flow

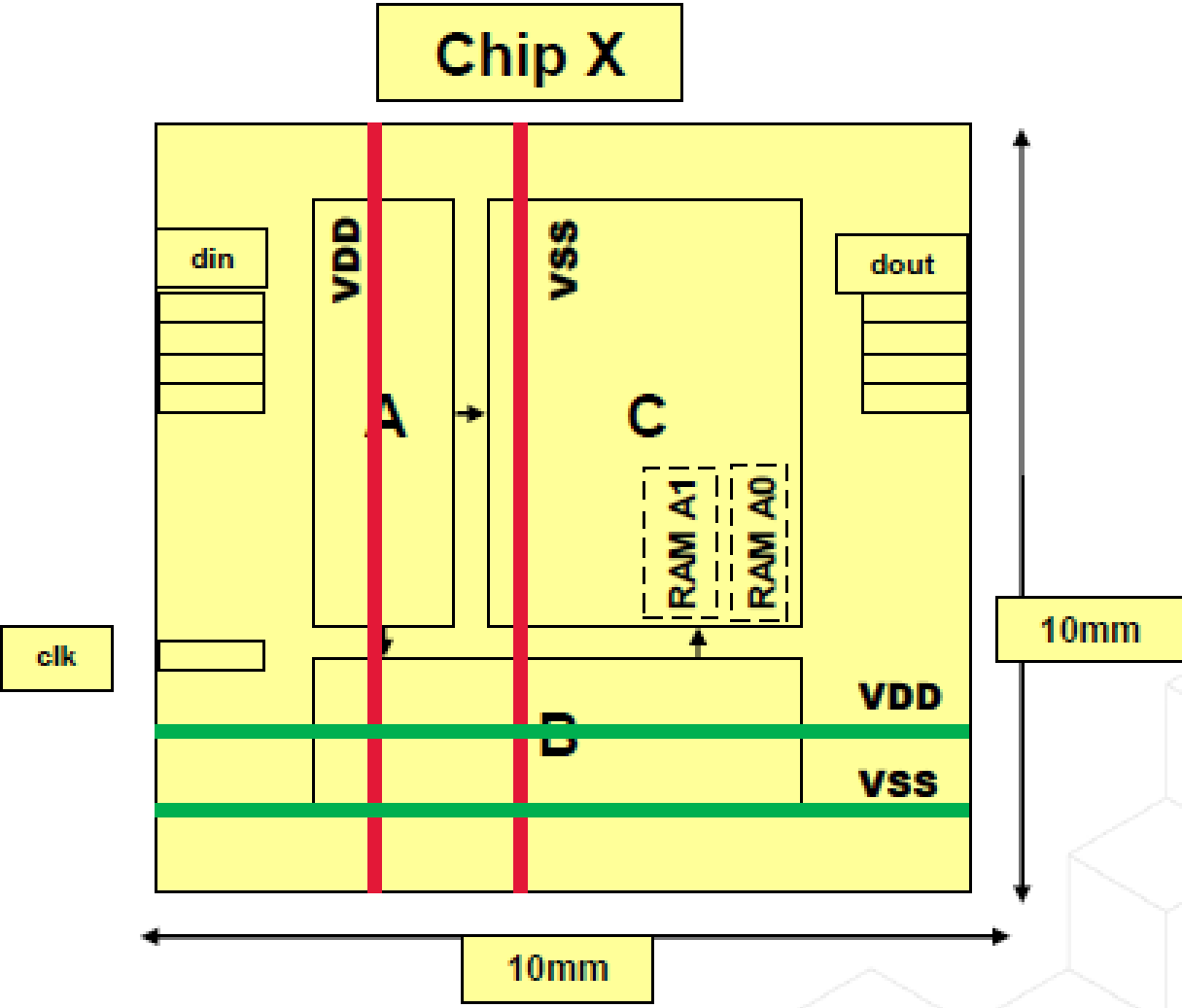


Input Files



Netlist	Design Netlist – Verilog or OpenAccess	Timing Constraints	Timing Constraint File(s) – SDCs for all modes
Floorplan File	Floorplan File – Innovus .fp file, DEF file	Timing Libraries	Timing Libraries – .libs for all sets, all Vts
Clock Tree Spec	Clock Tree Specification File (automatically generated from .sdc)	Physical Libraries	LEF Libraries or OpenAccess
Scan Information	Scan chain information – Tcl or DEF	Technology Files	Technology Files – Cap Tables or Quantus™ QRC
I/O Information	I/O Pad file (pads or pins)	Power Intent File	CPF or IEEE 1801 file
GDS Layer Map	GDS Layer Map File (if using GDS Format)		

Floor Plan

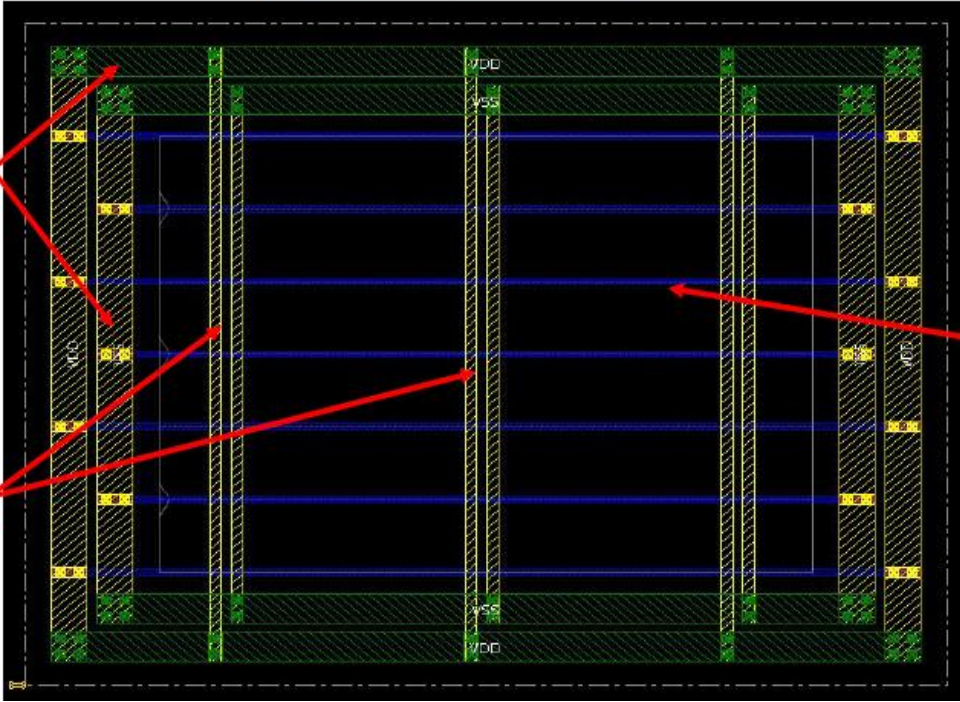


Power Plan

Snapshot of a power mesh

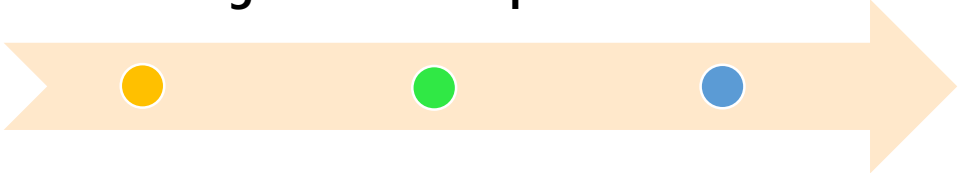
Power Rings

Power Stripes



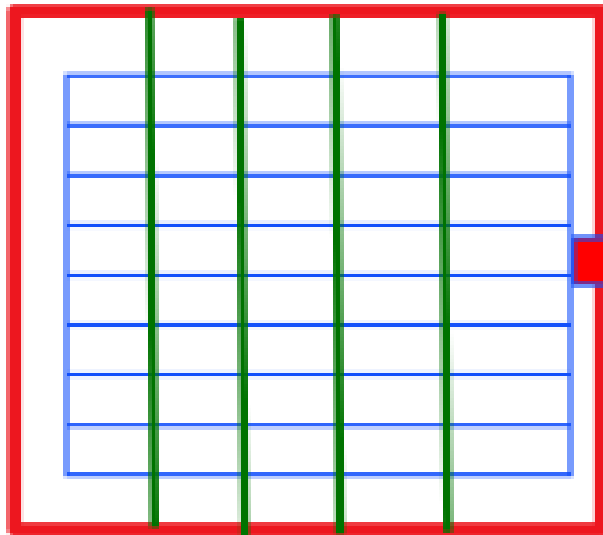
Power Rails

- 1. Power Ring
- 2. Power Stripe
- 3. Power Rail

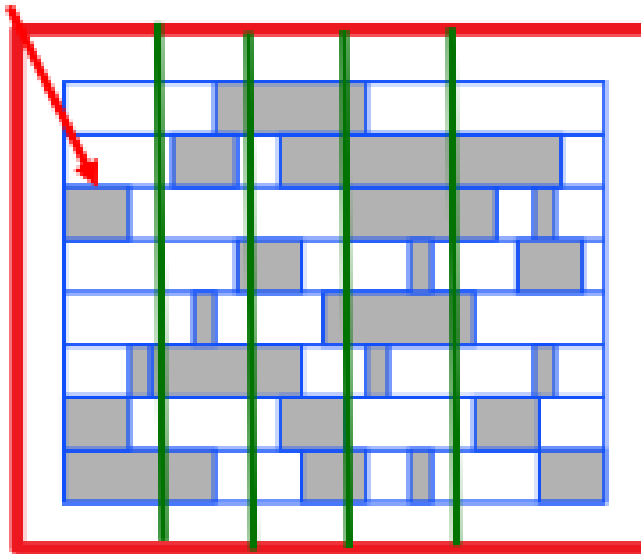
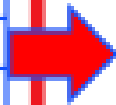


Placement

Standard Cell



Design Before Placement



Design After Placement

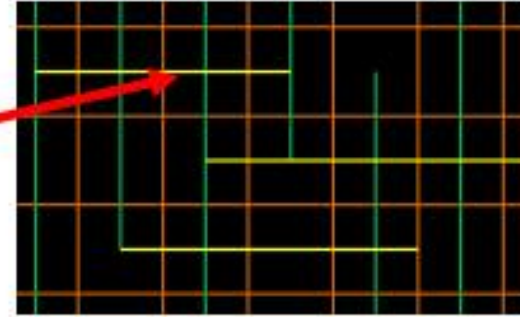
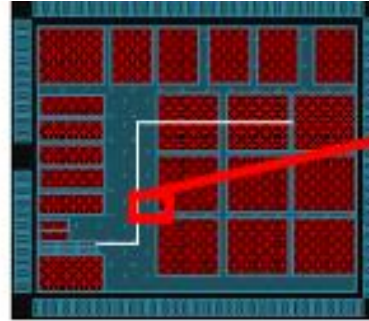
Sites in the LEF File

```
SITE coreSite  
  SYMMETRY y ;  
  CLASS core ;  
  SIZE 0.668 BY 5.040 ;  
END coreSite  
  
SITE pad  
  SYMMETRY x y r90 ;  
  CLASS pad ;  
  SIZE 0.100 BY 235.000 ;  
END pad  
  
SITE corner  
  SYMMETRY x y r90 ;  
  CLASS pad ;  
  SIZE 235.000 BY 235.000 ;  
END corner
```


Routing

Global Route

Design divided into Gcells and routes assigned to tracks
Congestion analyzed



Detail Route

Creates the routes
Applies all DRC rules
Gives priority to critical nets

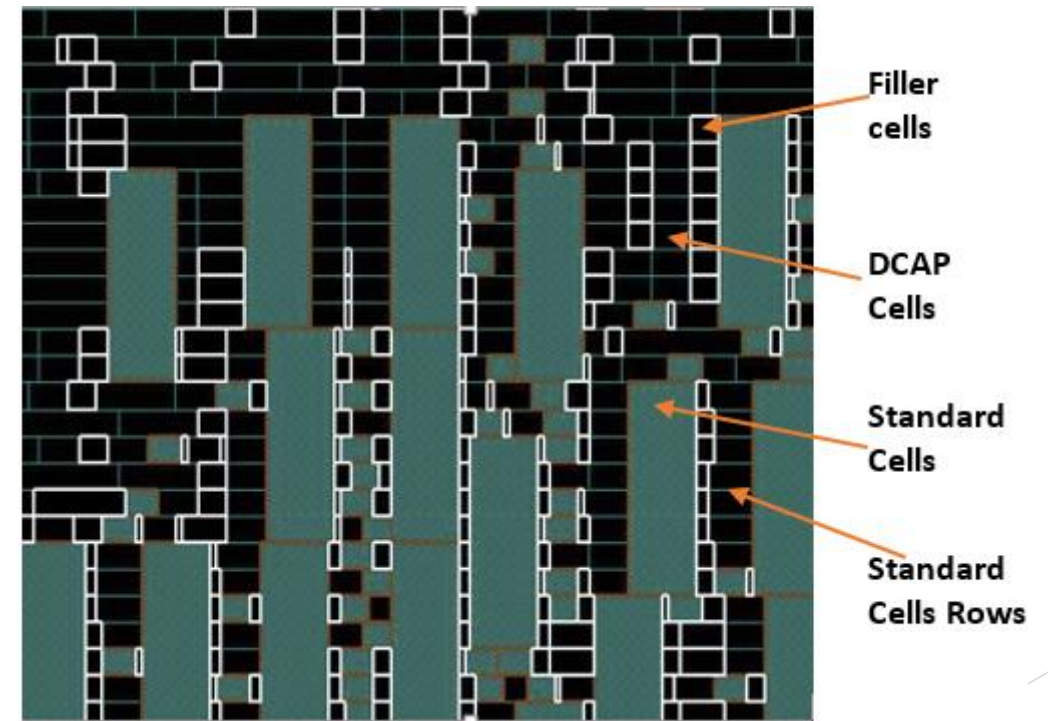
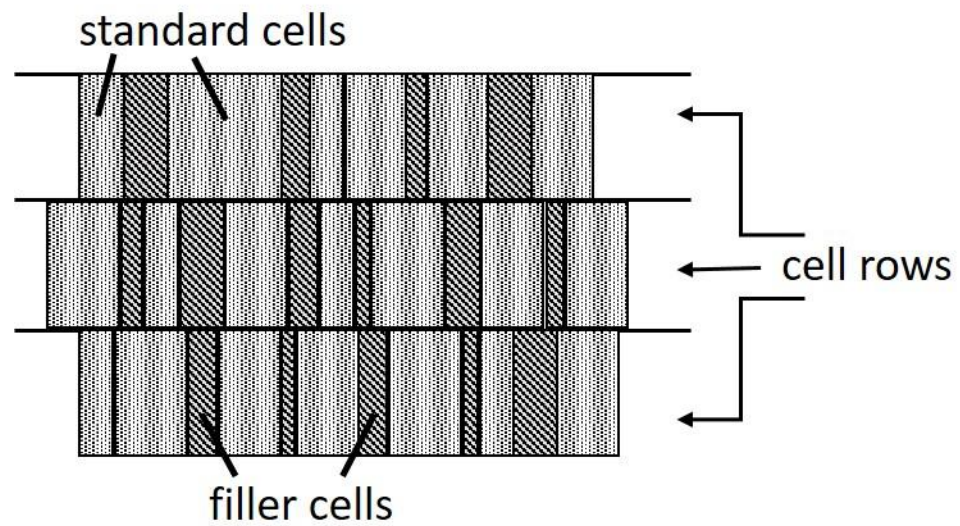


Search and Repair

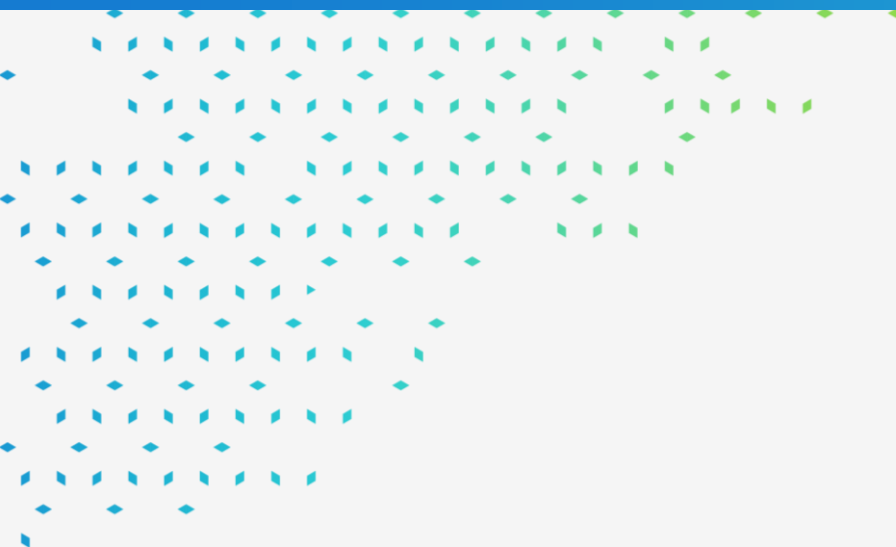
Surgical fixes for individual vias and wires
Search area increases by 3x3, 5x5, and 7x7 in Gcell areas
Deletion and reroute as a last resort if no other solution is found



Adding Filler and Chip Finish

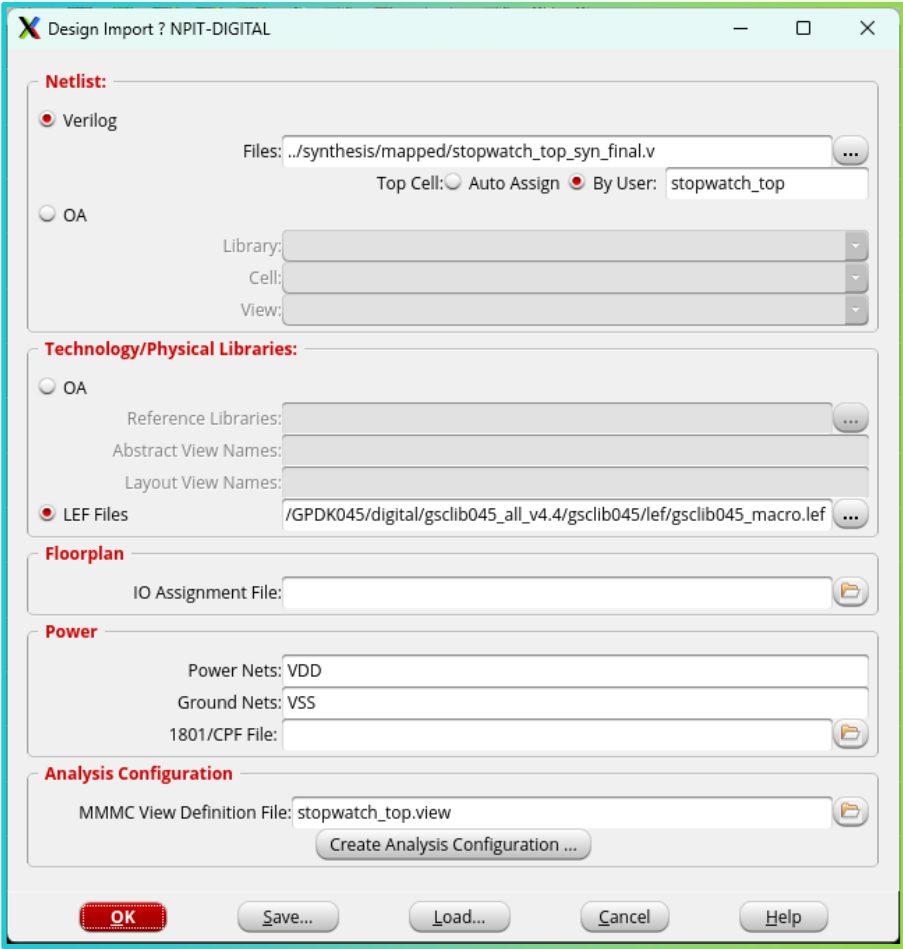
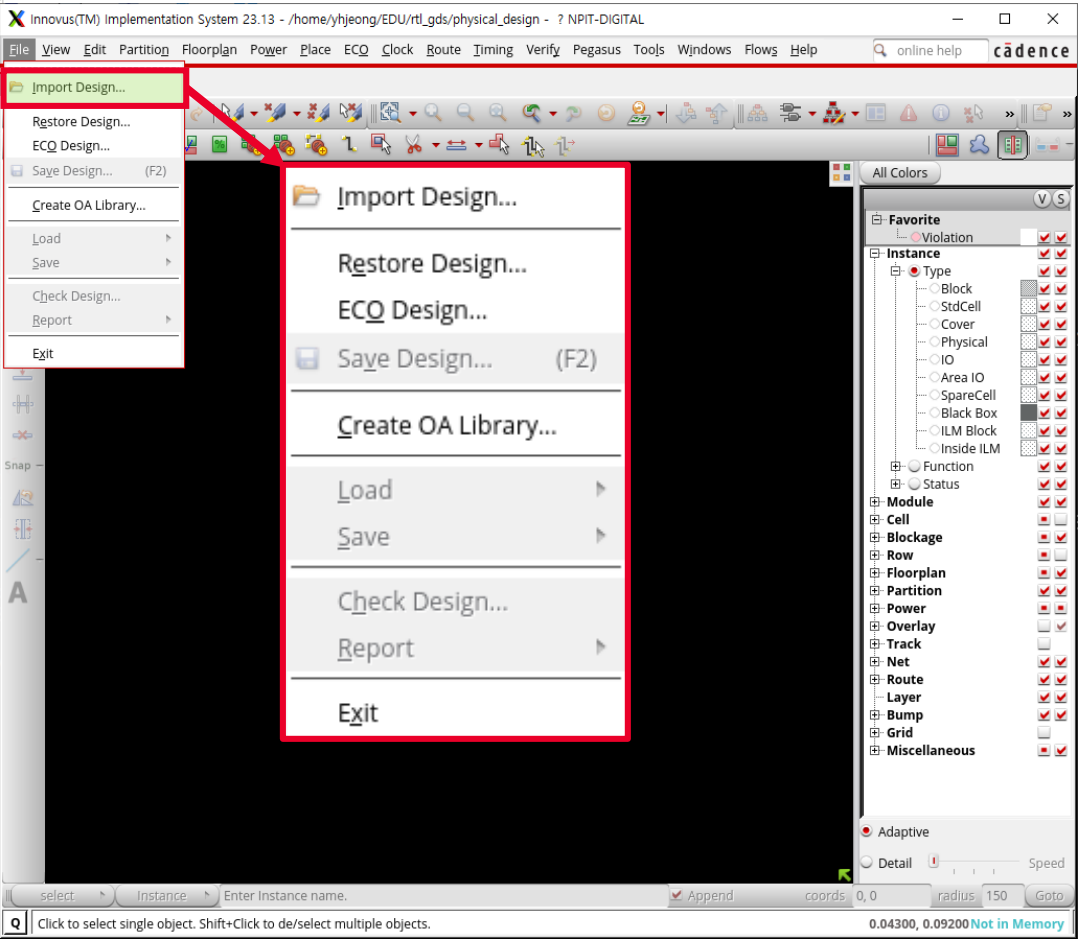


2. Interactive APR

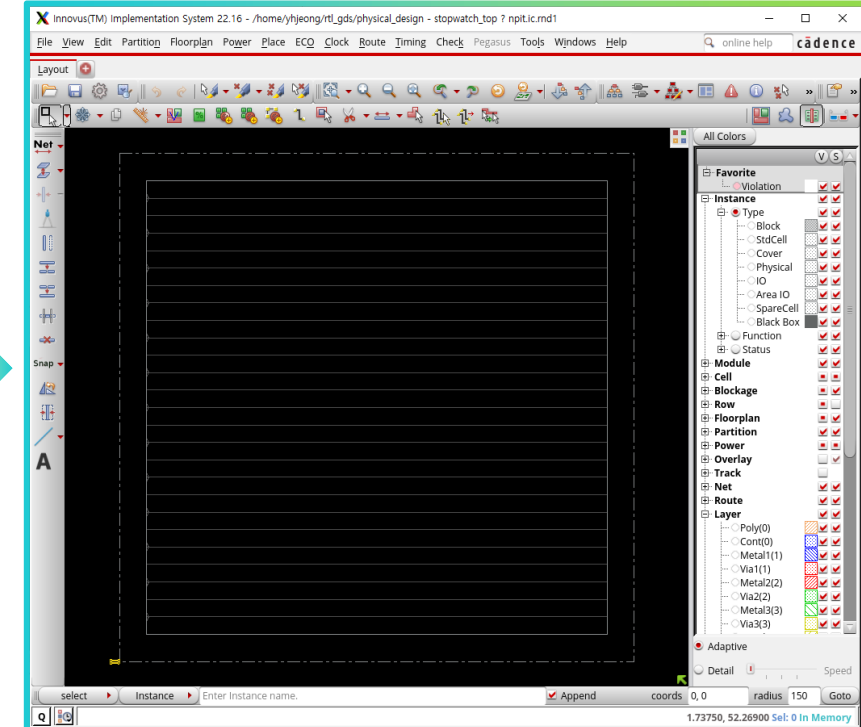
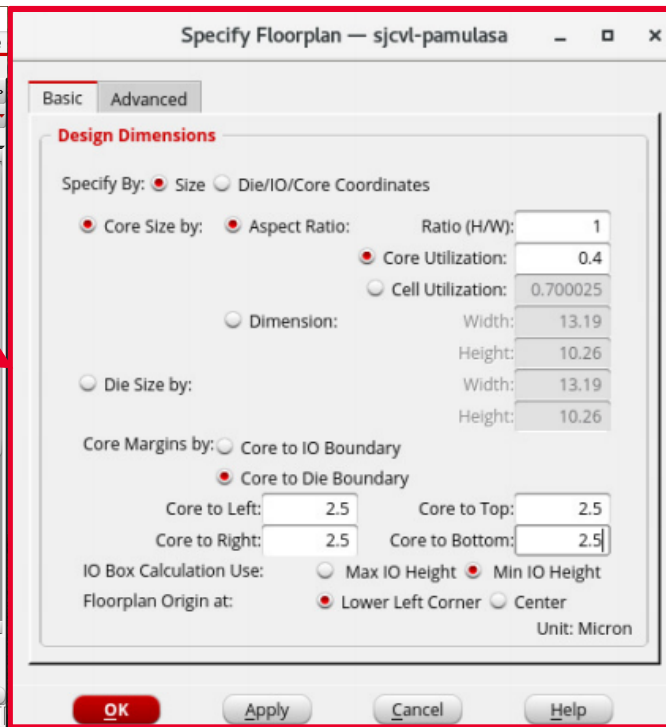
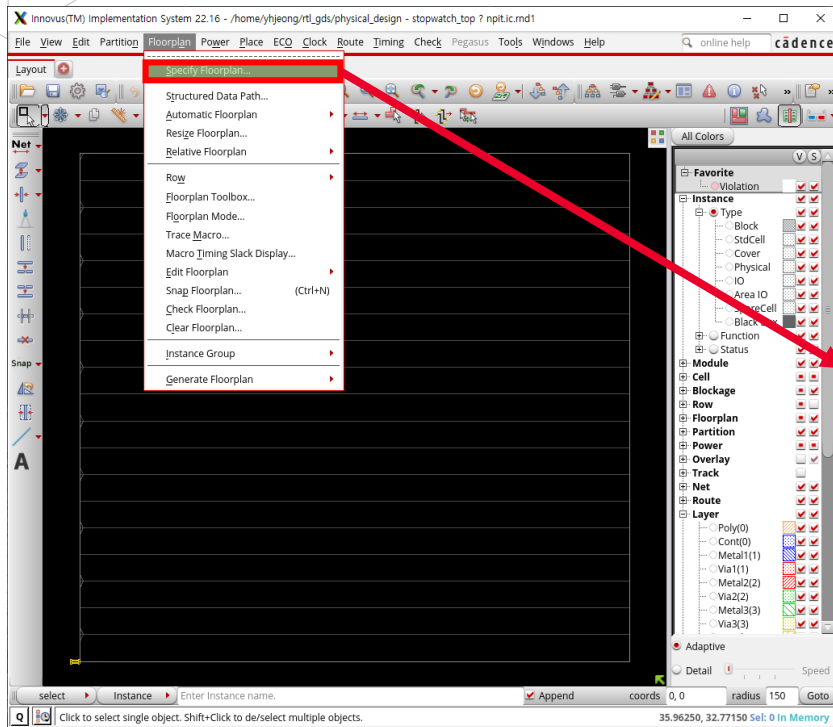


Init Design

Import Design



Specify Floorplan



Pin Location 작성(Optional)

```
(globals
  version = 3
  io_order = default
)

(iopin
  (top
    (pin name="clk"      offset=16      layer=2 width=0.0800 depth=0.2500 place_status=placed )
    (pin name="reset_n"  space=3.0      layer=2 width=0.0800 depth=0.2500 place_status=placed )
    (pin name="start"    space=3.0      layer=2 width=0.0800 depth=0.2500 place_status=placed )
    (pin name="stop"     space=3.0      layer=2 width=0.0800 depth=0.2500 place_status=placed )
  )

  (left
  )

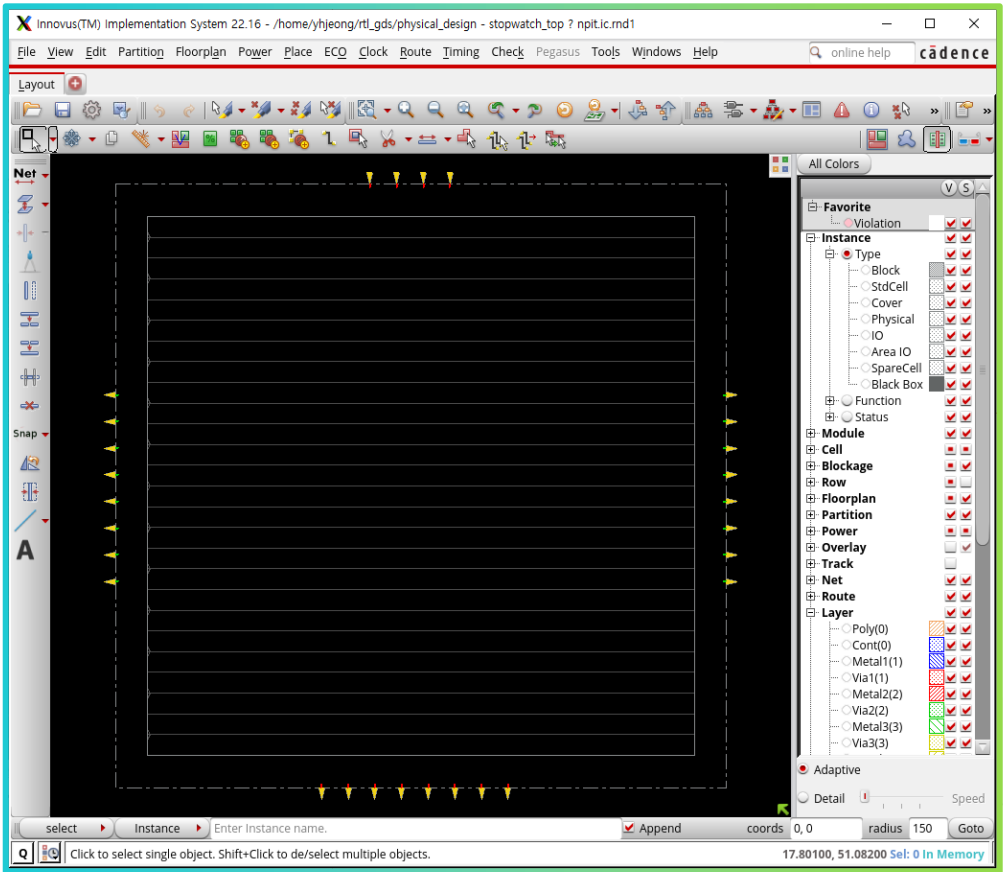
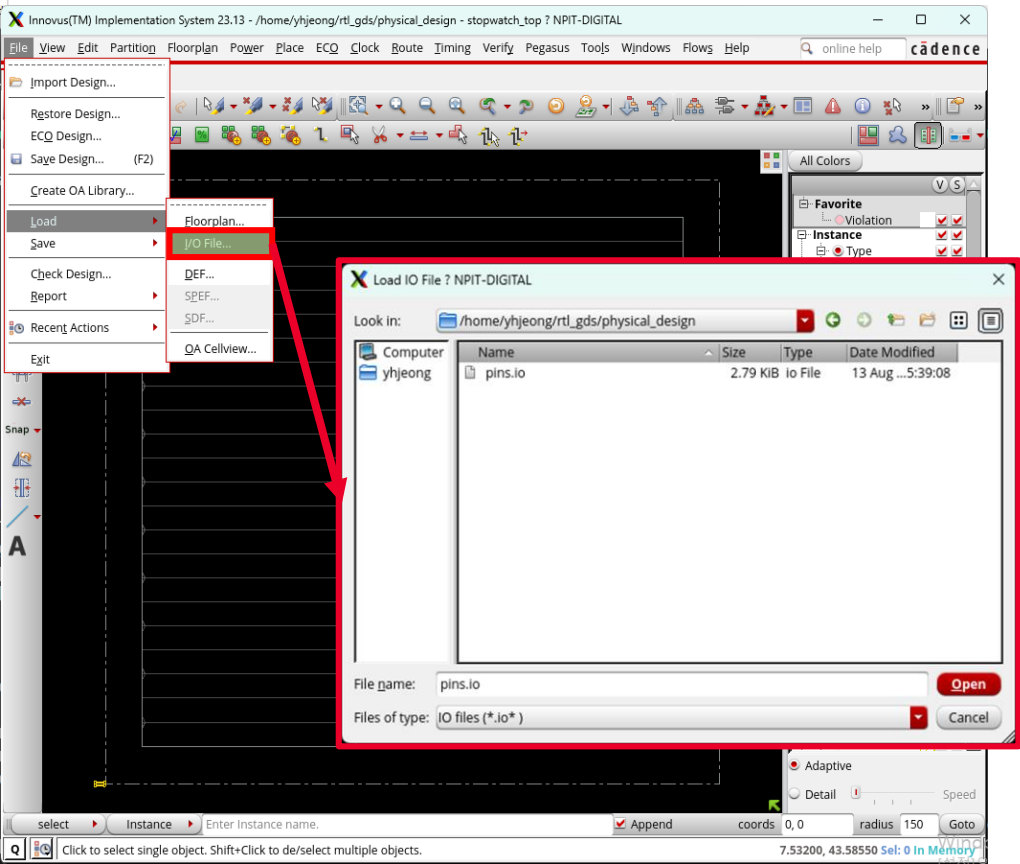
  (bottom
  )

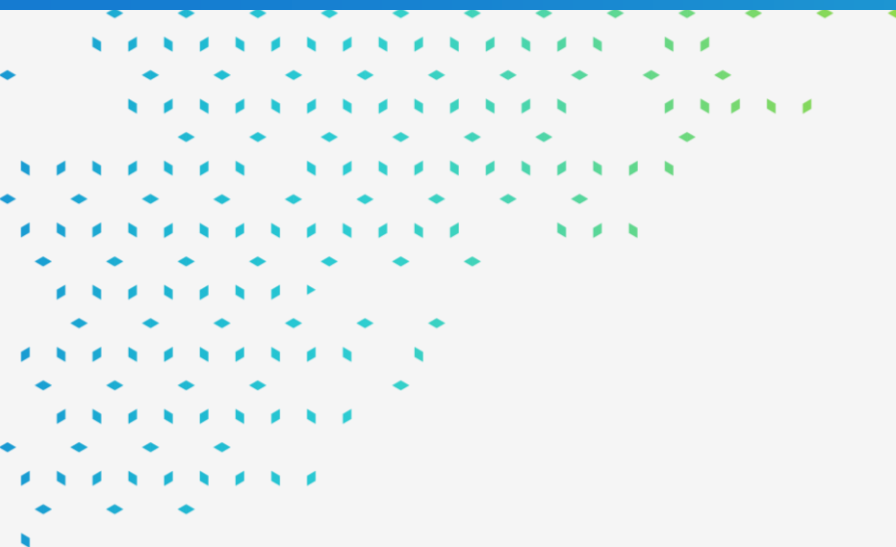
  (right
  )
)
```

```
OAI31X1 g993_5526(.A0 (i[0]), .A1 (i[2]), .A2 (n_0), .B0 (stop), .Y
(n_23));
NAND2BX1 g994_6783(.AN (n_5), .B (stop), .Y (n_22));
ADDHX1 g995_3680(.A (g[1]), .B (g[0]), .CO (n_17), .S (n_18));
ADDHX1 g996_1617(.A (h[1]), .B (h[0]), .CO (n_15), .S (n_16));
ADDHX1 g997_2802(.A (i[1]), .B (i[0]), .CO (n_13), .S (n_14));
ADDHX1 g998_1705(.A (e[1]), .B (e[0]), .CO (n_11), .S (n_12));
ADDHX1 g999_5122(.A (d[1]), .B (d[0]), .CO (n_9), .S (n_10));
ADDHX1 g1000_8246(.A (f[1]), .B (f[0]), .CO (n_7), .S (n_8));
NOR4BBX1 g1001_7098(.AN (f[3]), .BN (f[1]), .C (f[2]), .D (f[0]), .Y
(n_20));
NOR4BBX1 g1002_6131(.AN (g[3]), .BN (g[1]), .C (g[2]), .D (g[0]), .Y
(n_19));
NOR4BBX1 g1003_1881(.AN (h[3]), .BN (h[1]), .C (h[2]), .D (h[0]), .Y
(n_6));
NOR4BBX1 g1004_5115(.AN (e[3]), .BN (e[1]), .C (e[2]), .D (e[0]), .Y
(n_5));
NOR4BBX1 g1005_7482(.AN (d[3]), .BN (d[1]), .C (d[2]), .D (d[0]), .Y
(n_4));
NAND4BX1 g1006_4733(.AN (sec_count[9]), .B (sec_count[2]), .C
(sec_count[8]), .D (sec_count[3]), .Y (n_3));
NOR4BX1 g1007_6161(.AN (sec_count[15]), .B (sec_count[11]), .C
(sec_count[10]), .D (sec_count[14]), .Y (n_2));
NOR4BX1 g1008_9315(.AN (sec_count[13]), .B (sec_count[7]), .C
(sec_count[6]), .D (sec_count[12]), .Y (n_1));
NAND2X1 g1009_9945(.A (i[1]), .B (i[3]), .Y (n_0));
endmodule

module stopwatch_top(clk, reset_n, start, stop, d, e, f, g, h, i);
  input clk, reset_n, start, stop;
  output [3:0] d, e, f, g, h, i;
  wire clk, reset_n, start, stop;
  wire [3:0] d, e, f, g, h, i;
  wire [1:0] en;
  wire [18:0] sec_count;
  sec_counter sec_counter(.clk (clk), .reset_n (reset_n), .en (en),
    .sec_count (sec_count));
  stopwatch_fsm stopwatch_fsm(.clk (clk), .reset_n (reset_n), .start
    (start), .stop (stop), .en (en));
  timer timer(.clk (clk), .reset_n (reset_n), .sec_count (sec_count),
    .stop (stop), .d (d), .e (e), .f (f), .g (g), .h (h), .i (i));
endmodule
```

Load IO(Optional)

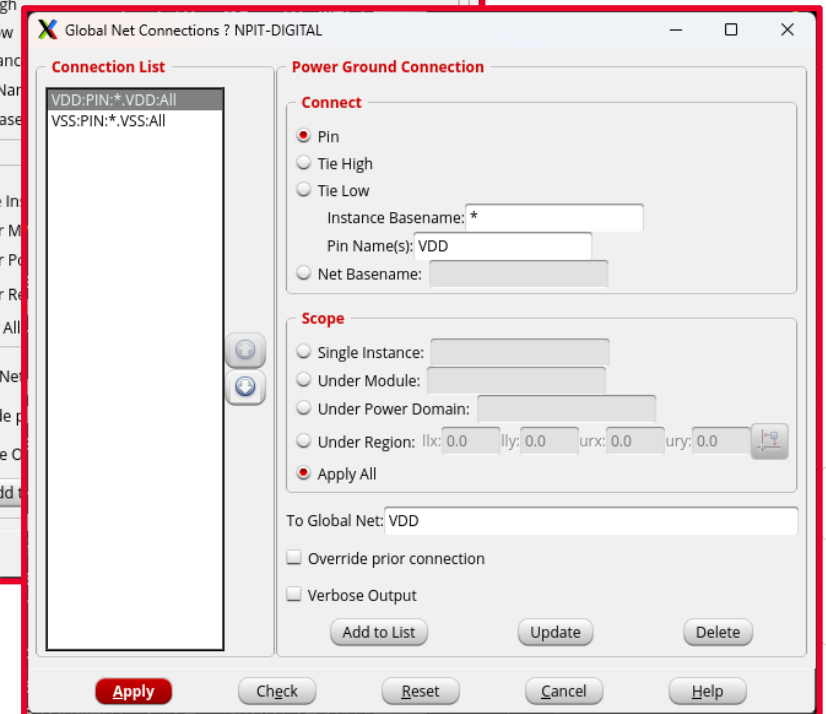
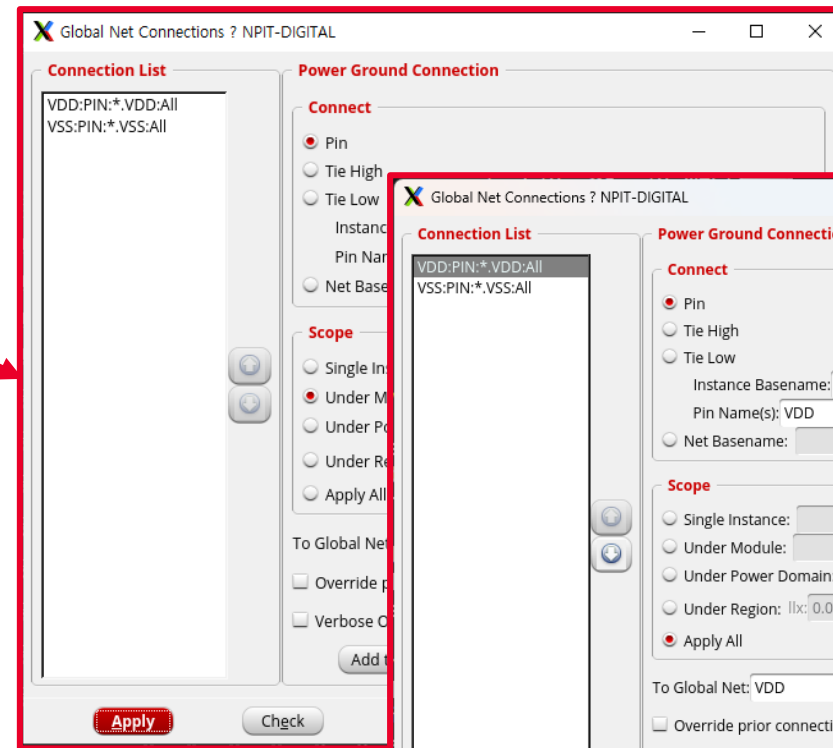
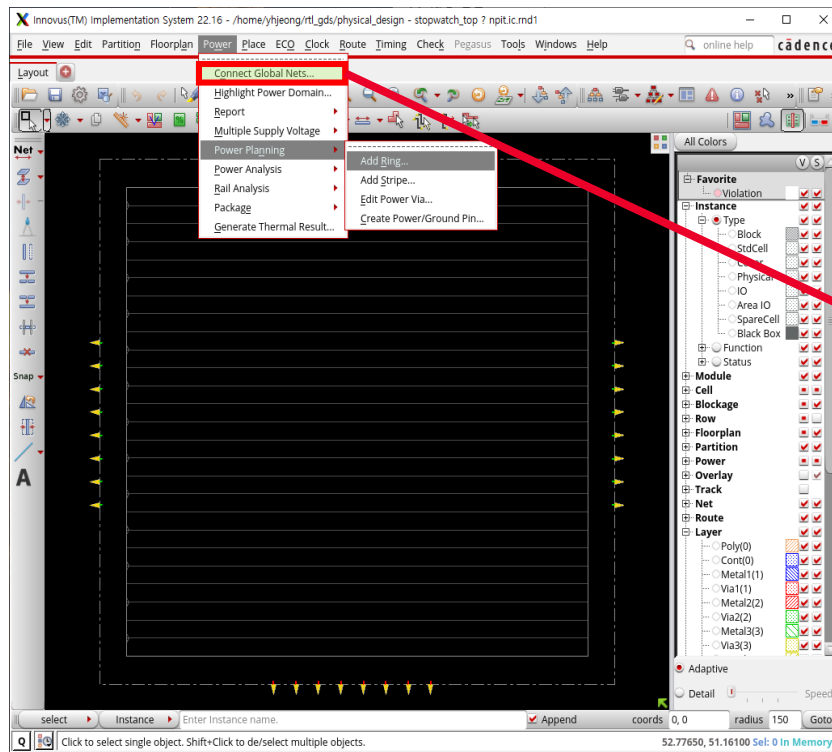




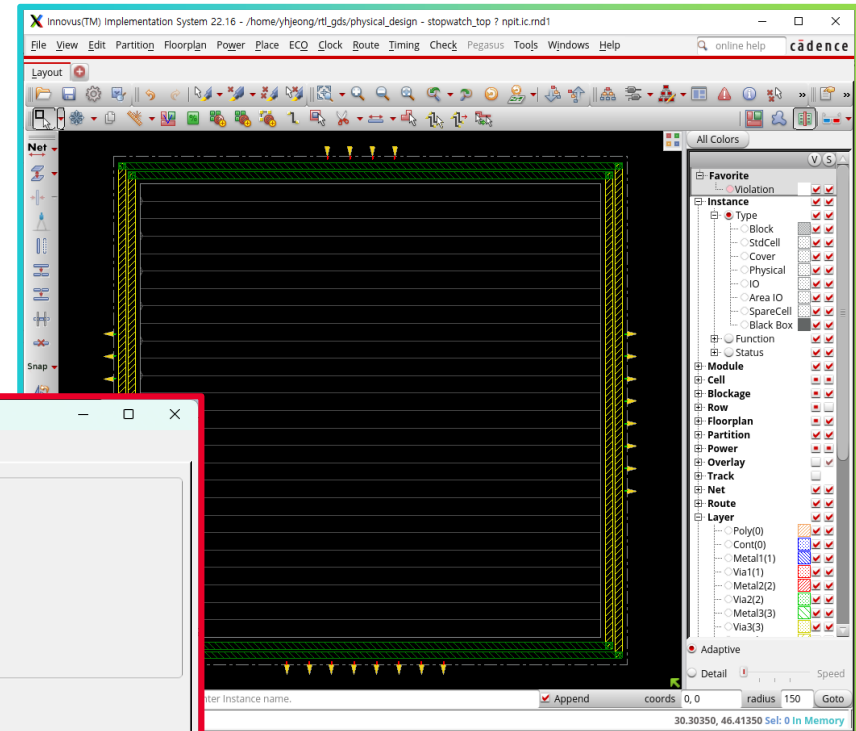
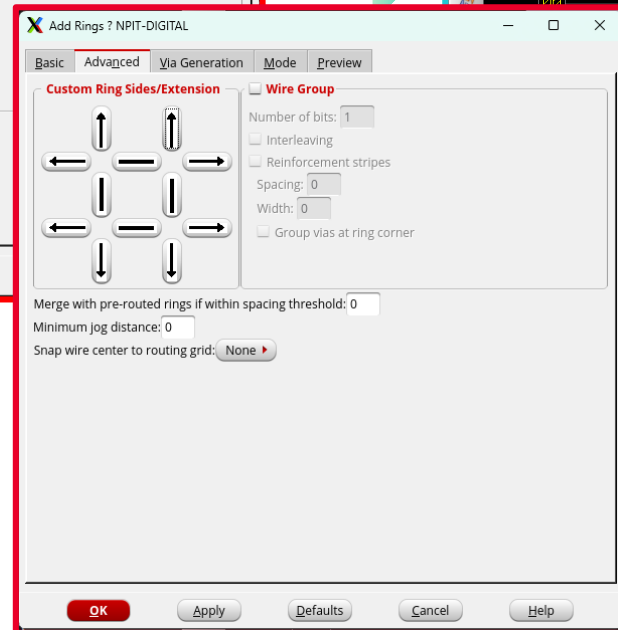
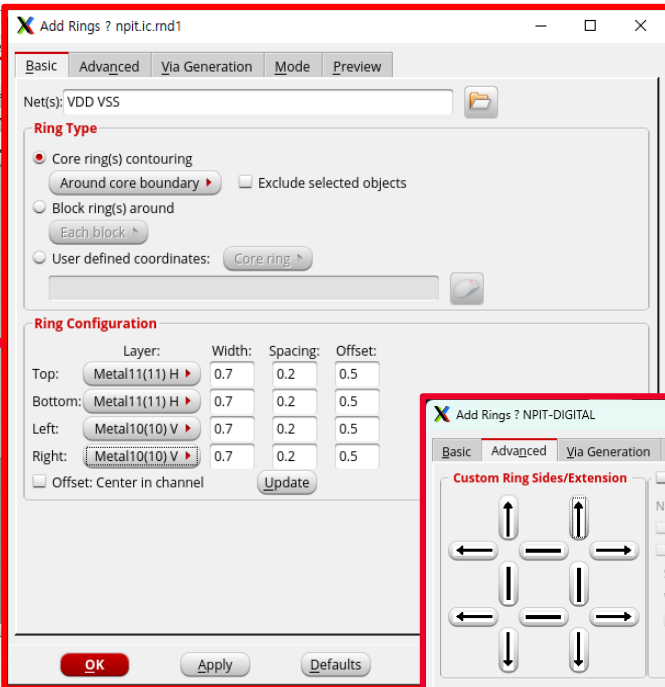
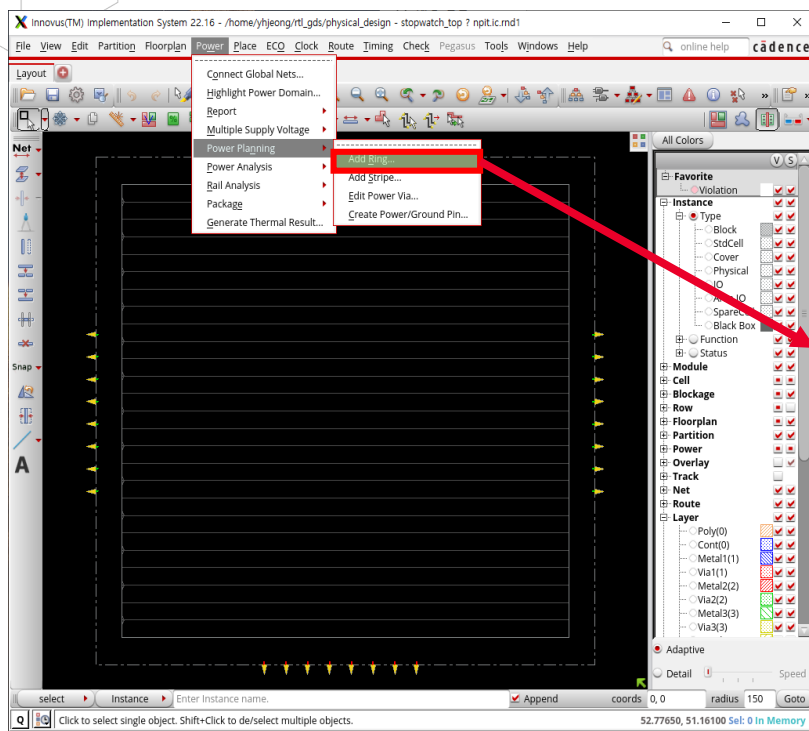
Power Plan

Connect Global Nets

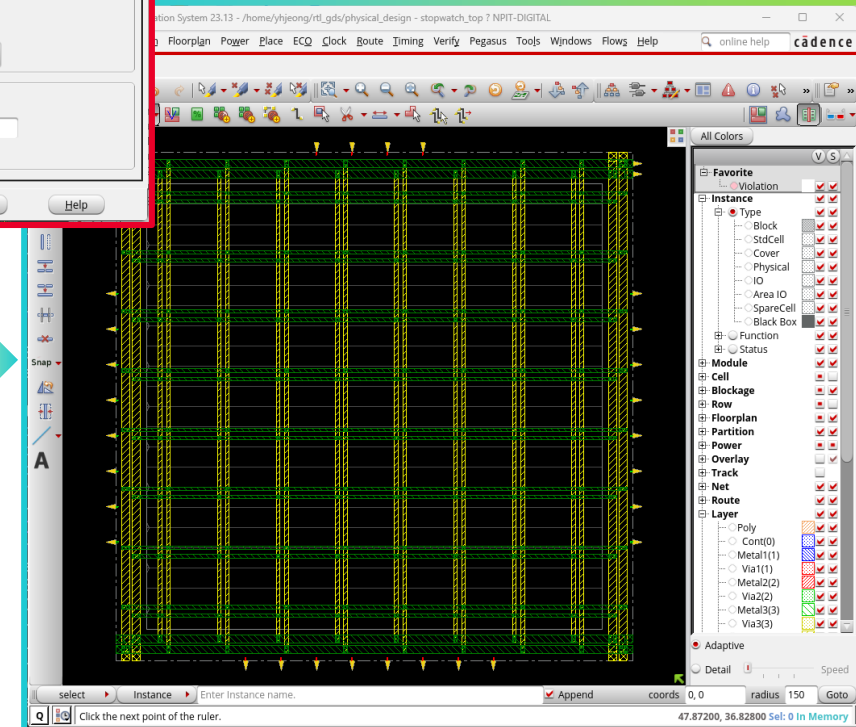
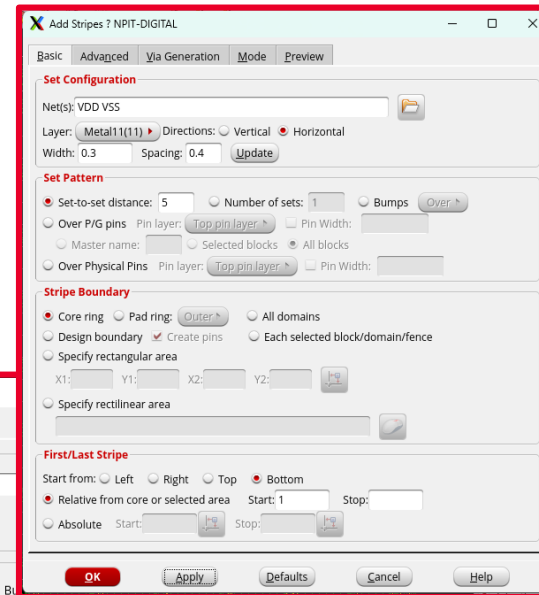
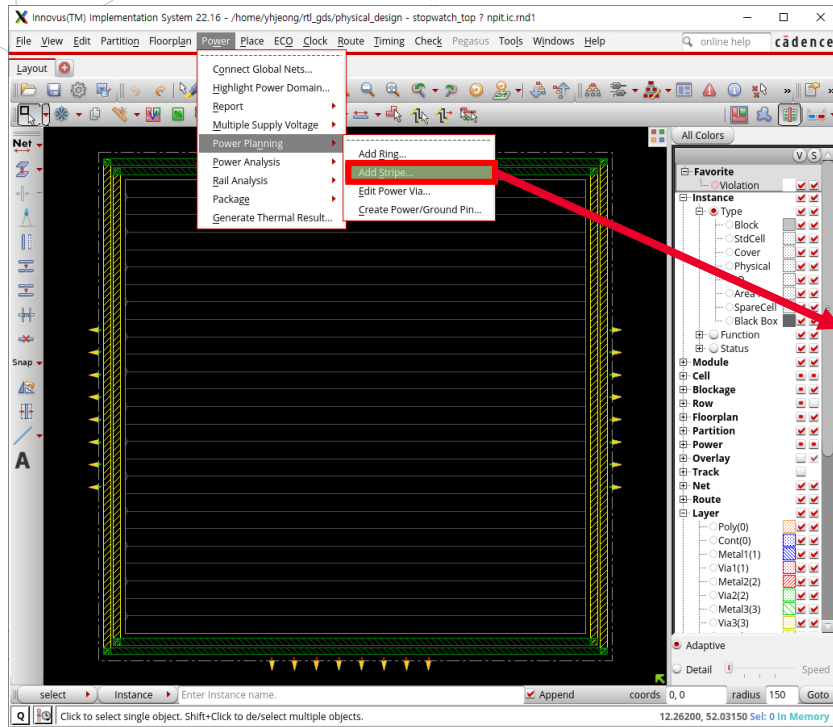
```
@innovus 3> connect_global_net VDD -type pg_pin -pin VDD -inst_base_name *  
@innovus 4> connect_global_net VSS -type pg_pin -pin VSS -inst_base_name *
```

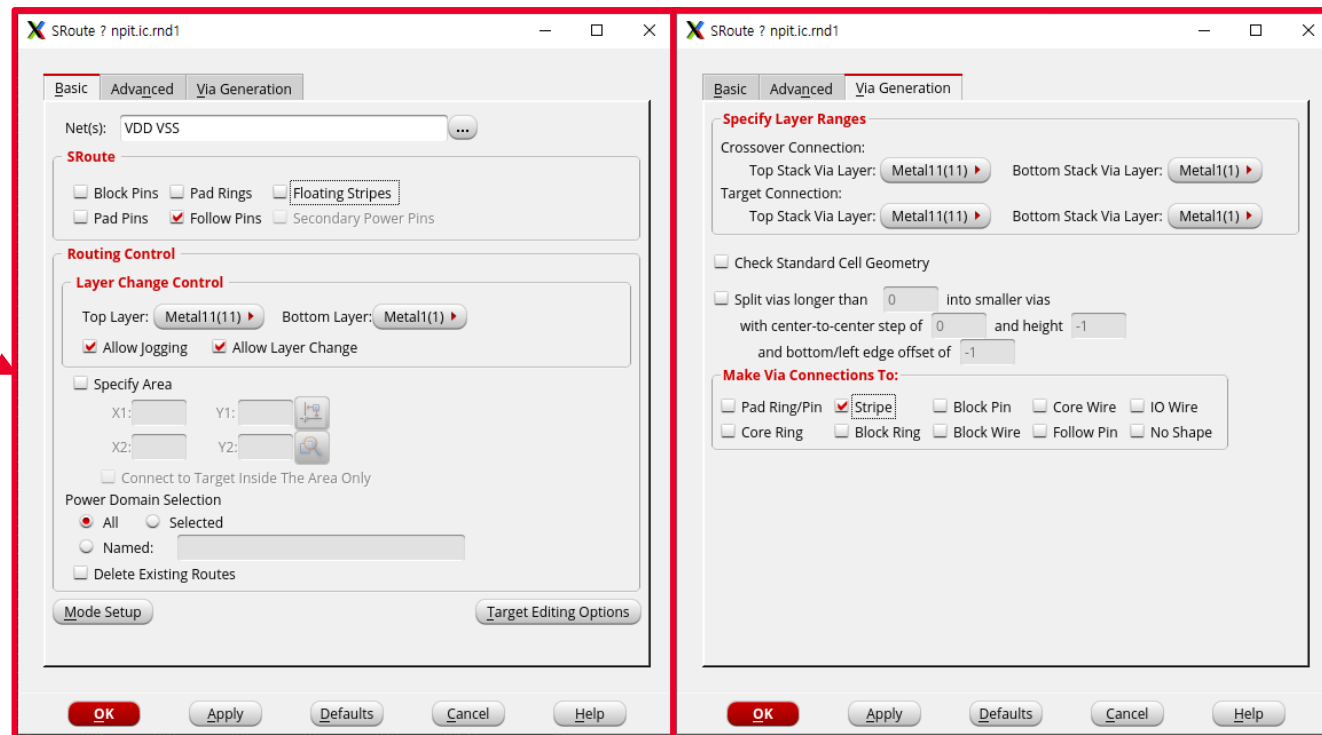
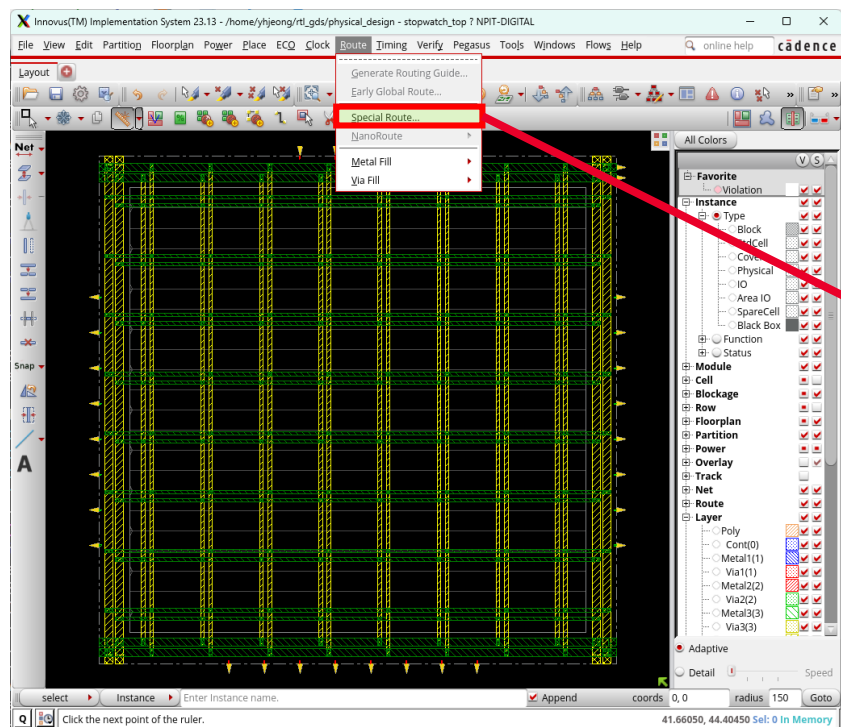


Add R

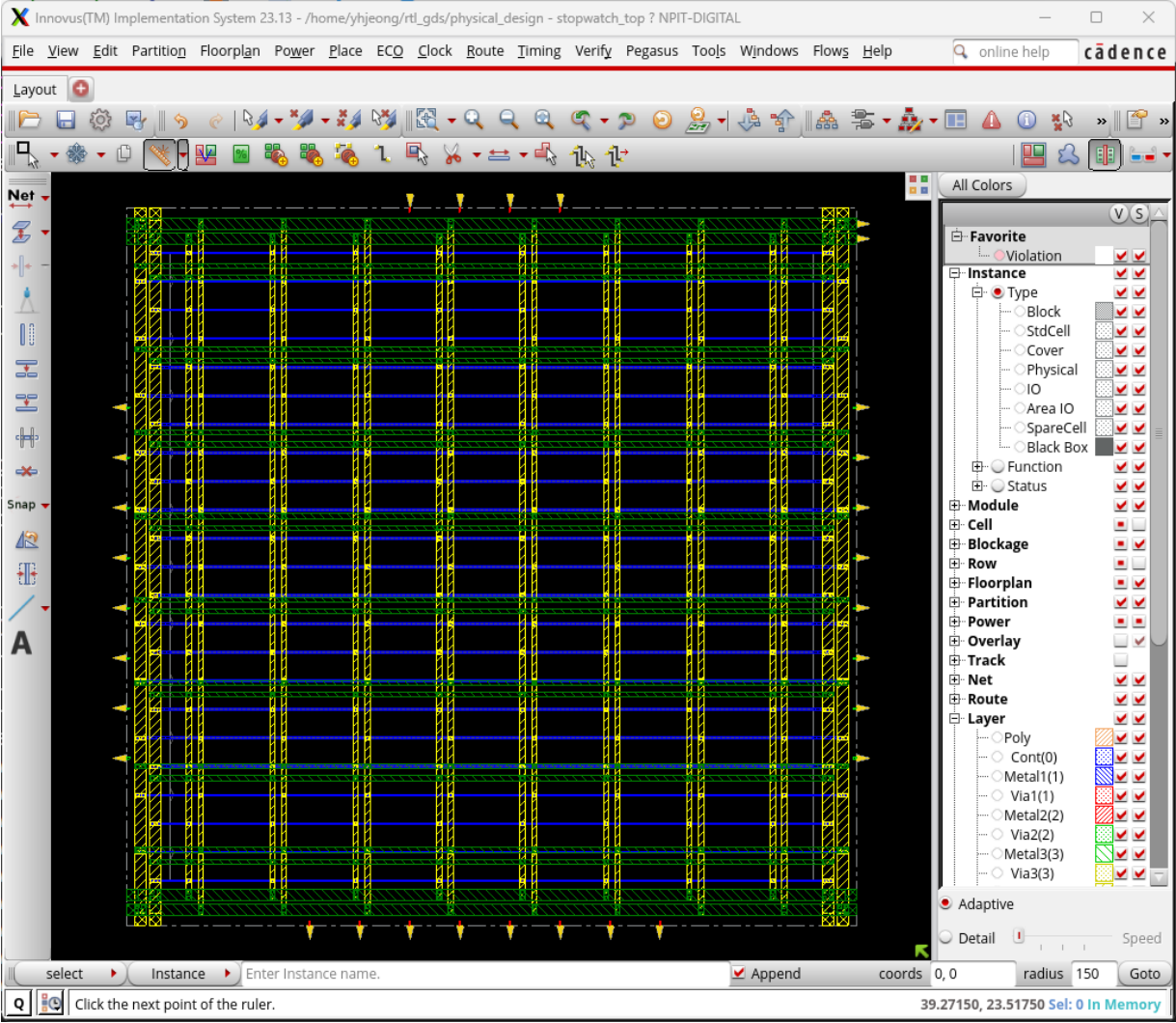


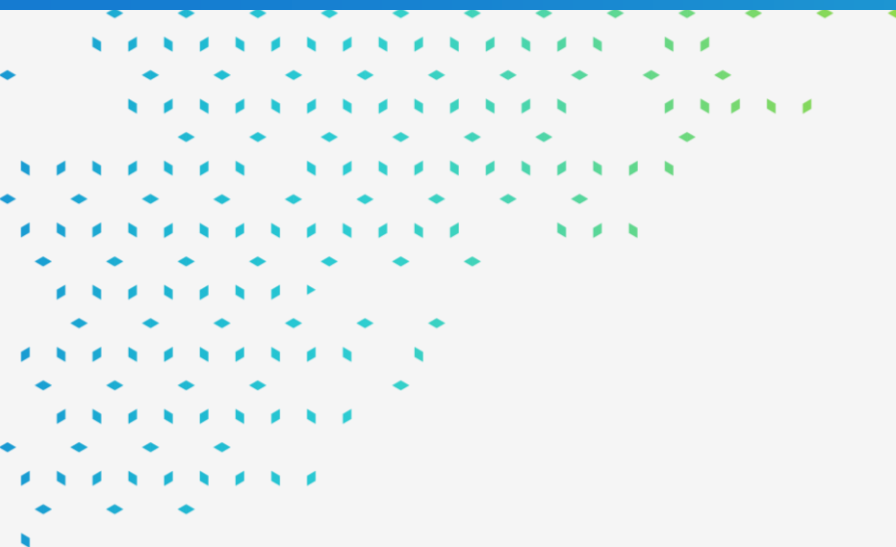
Add Stripes





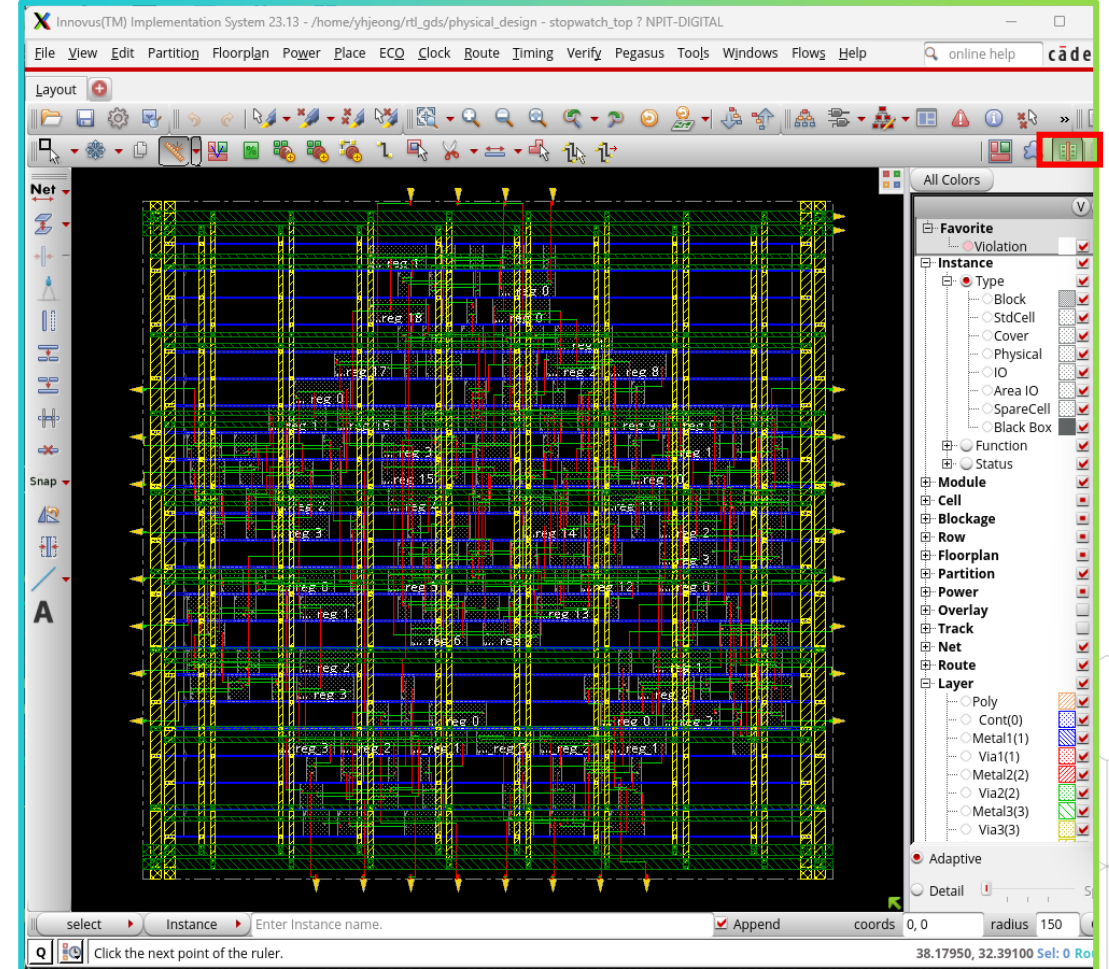
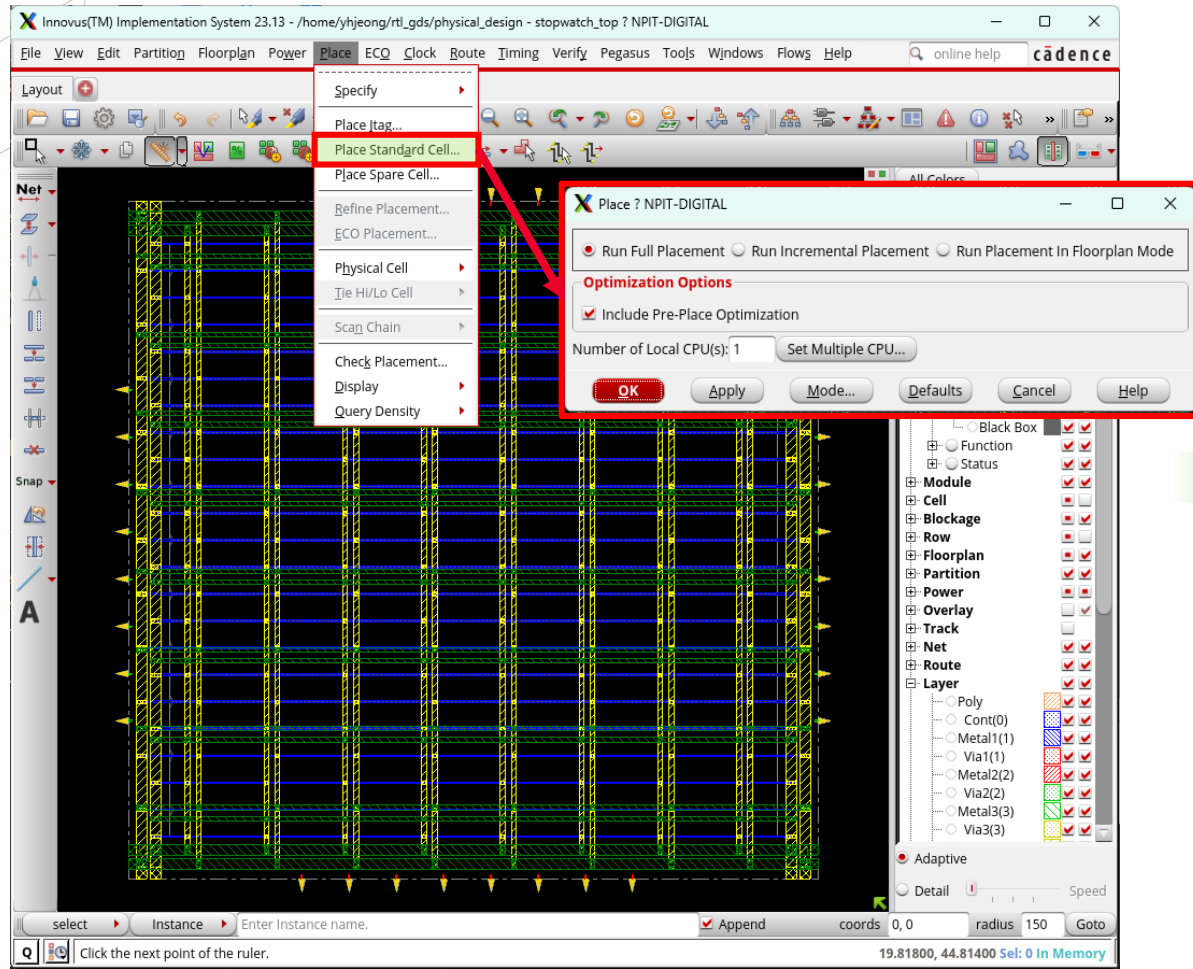
Result



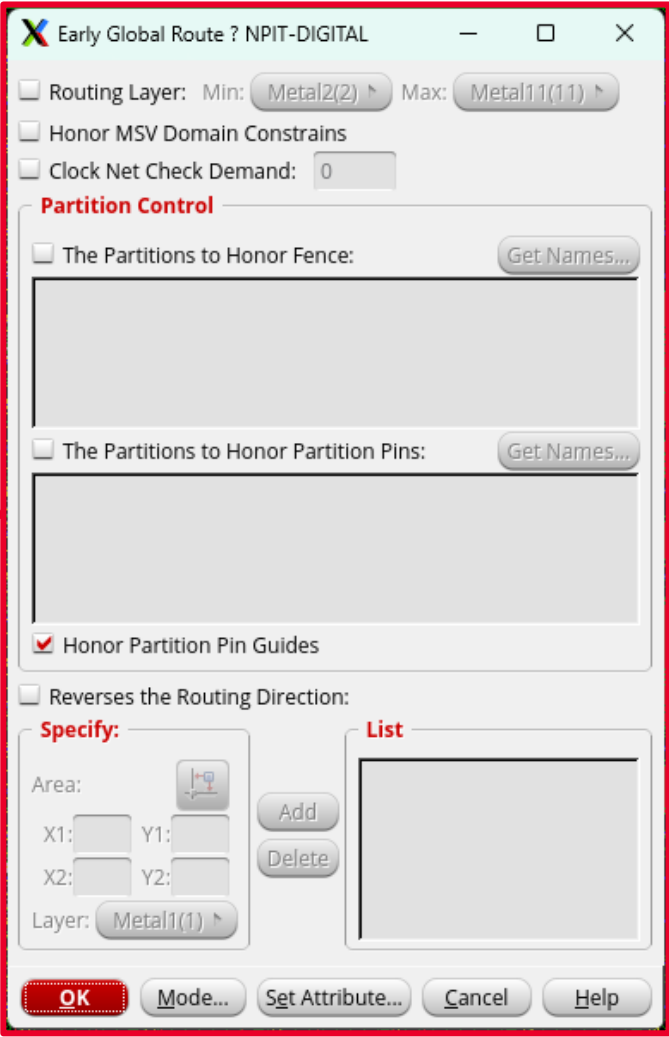
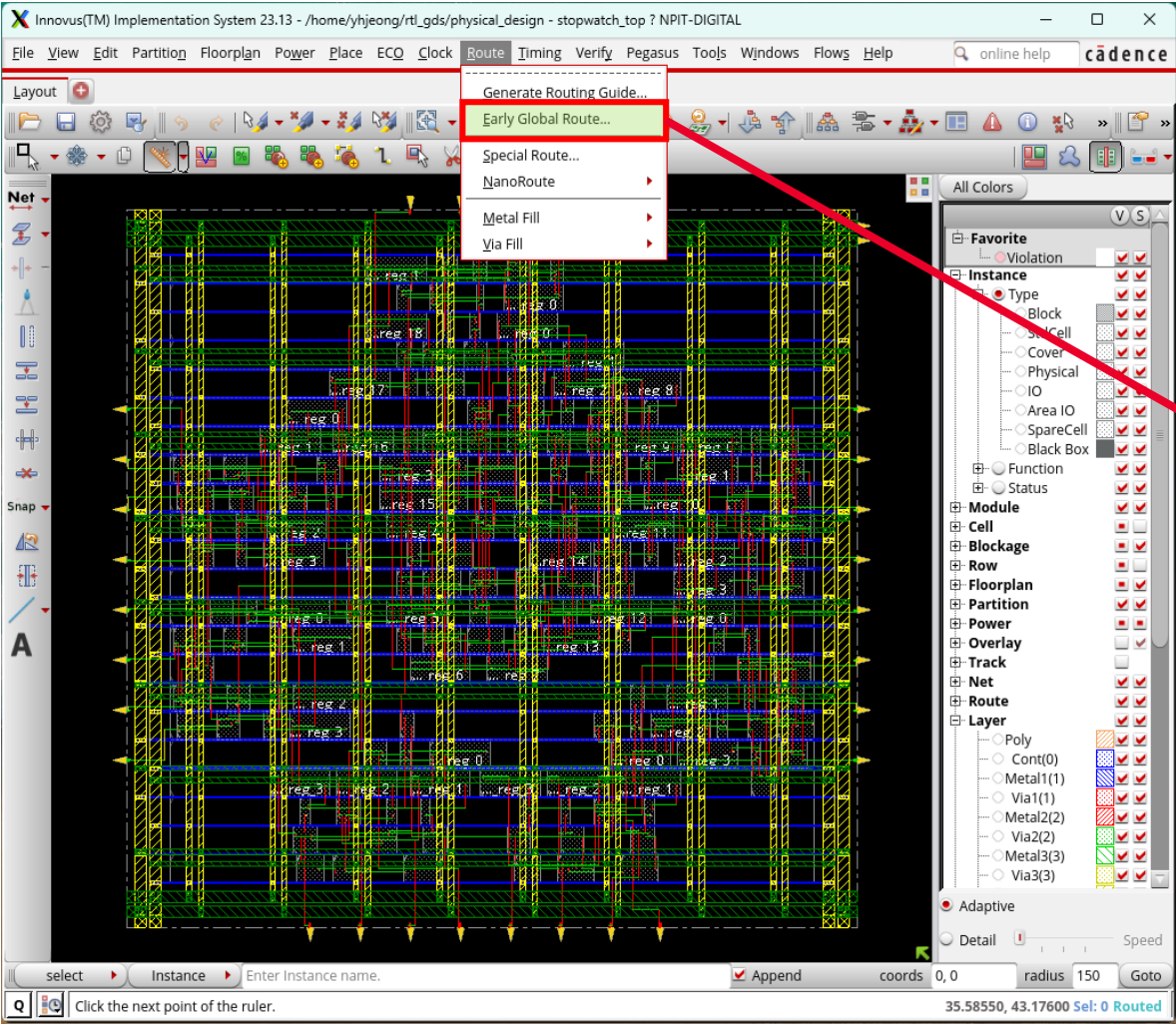


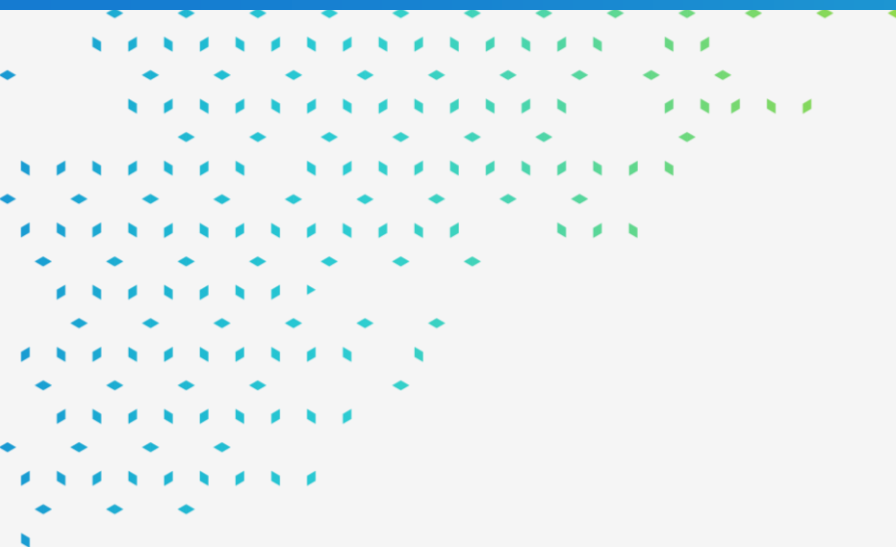
Placement

Place



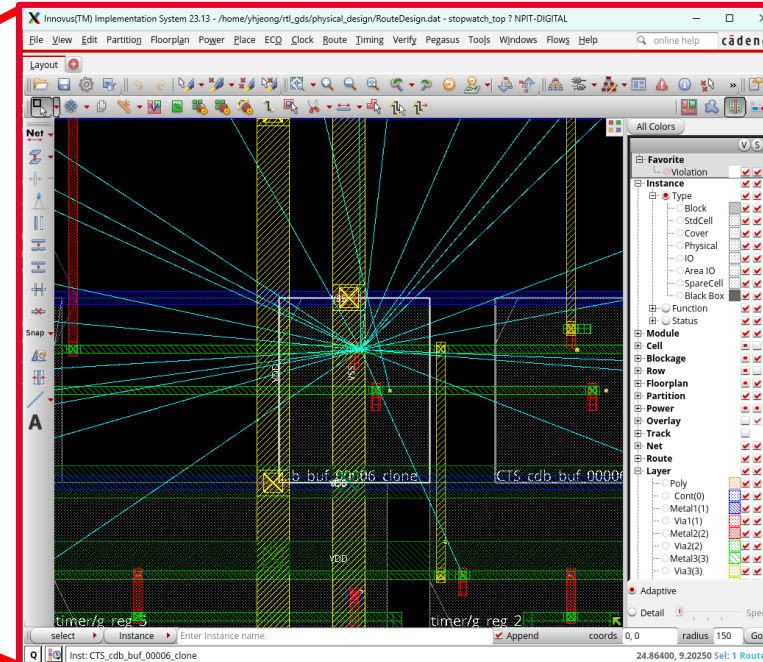
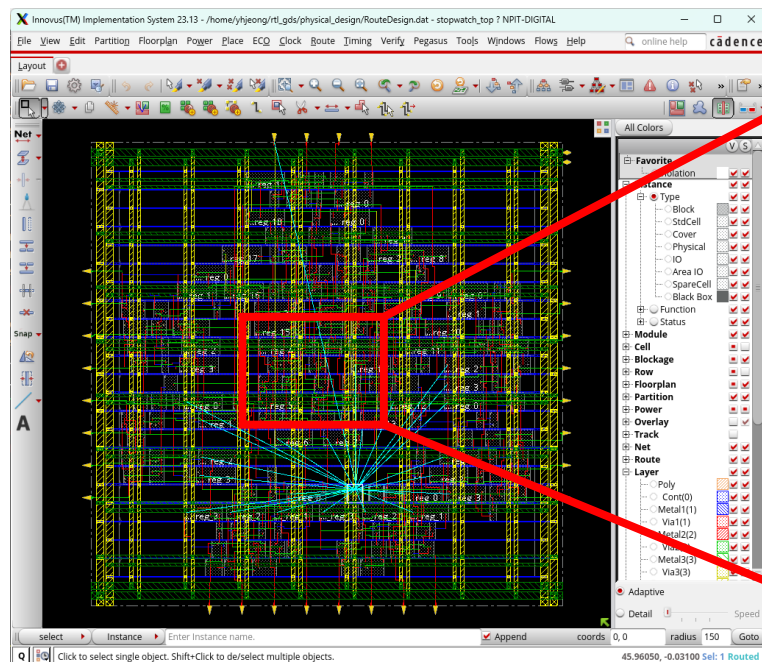
Early Global Route





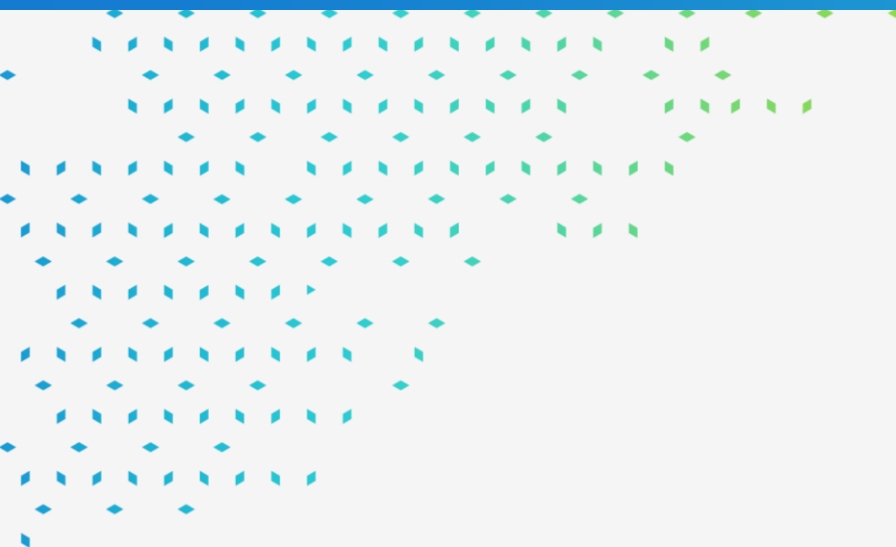
Clock Tree Synthesis(CTS)





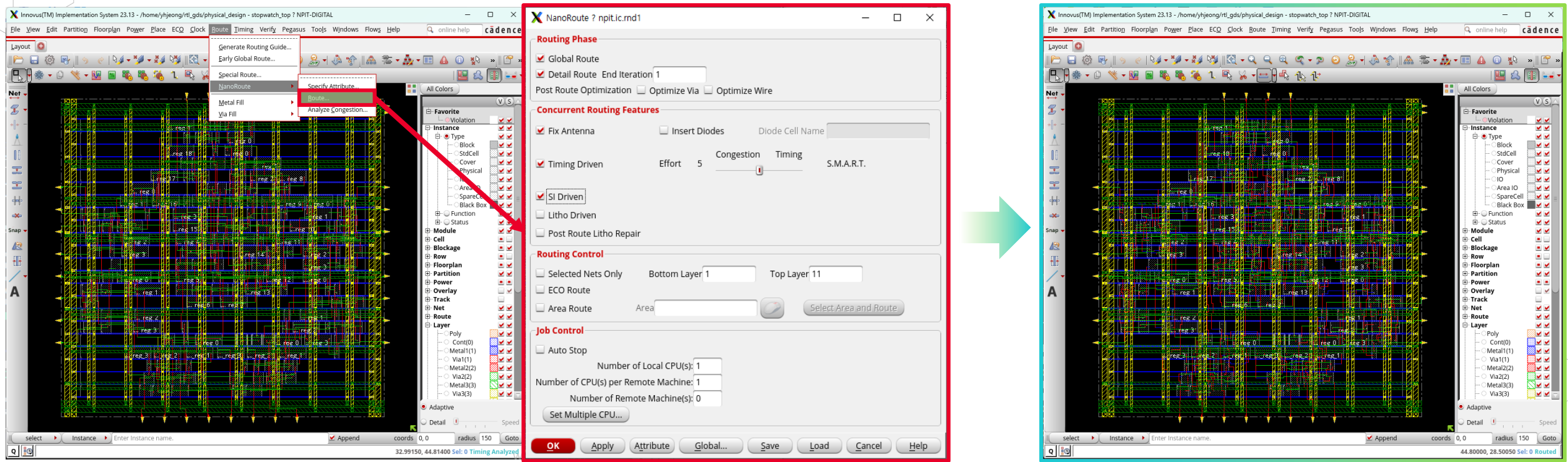
순서대로 입력

```
create_clock_tree_spec
ccopt_design
set_interactive_constraint_modes [all_constraint_modes -active]
set_propagated_clock [all_clocks]
```

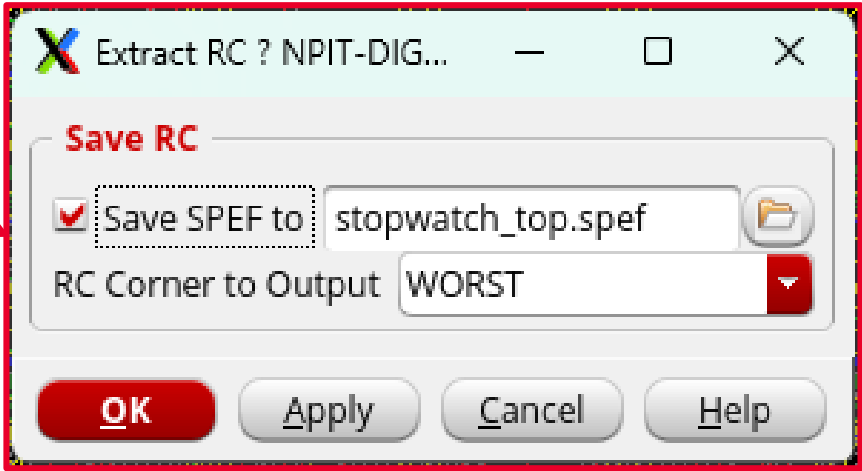
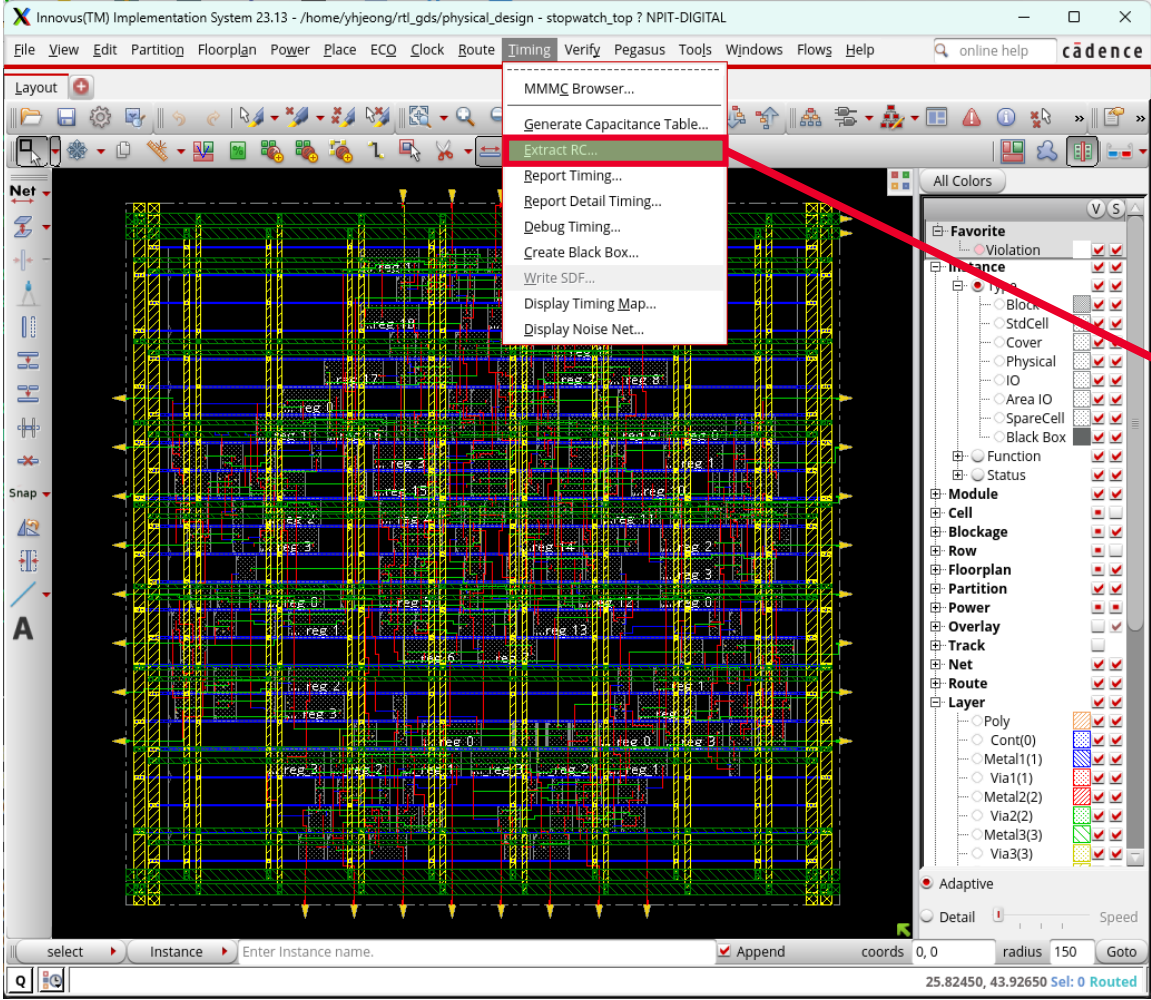



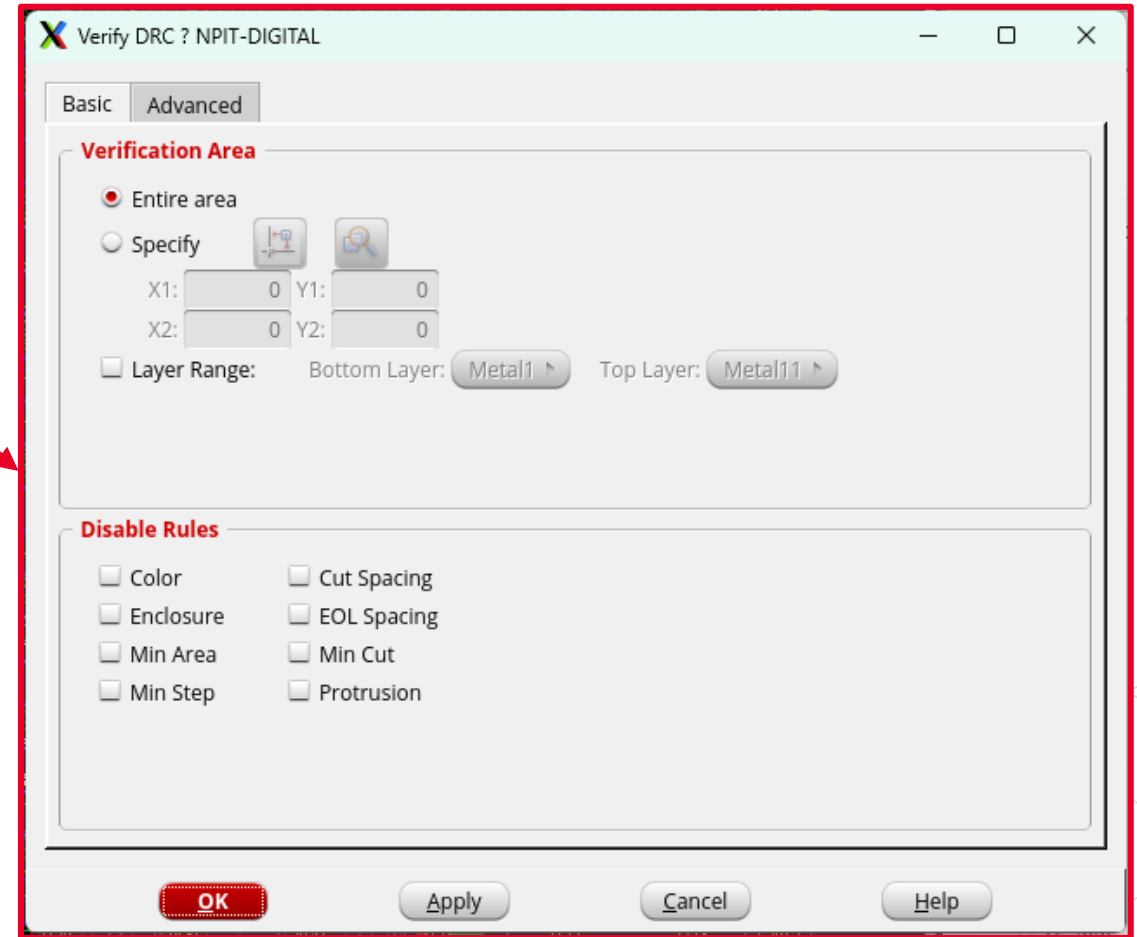
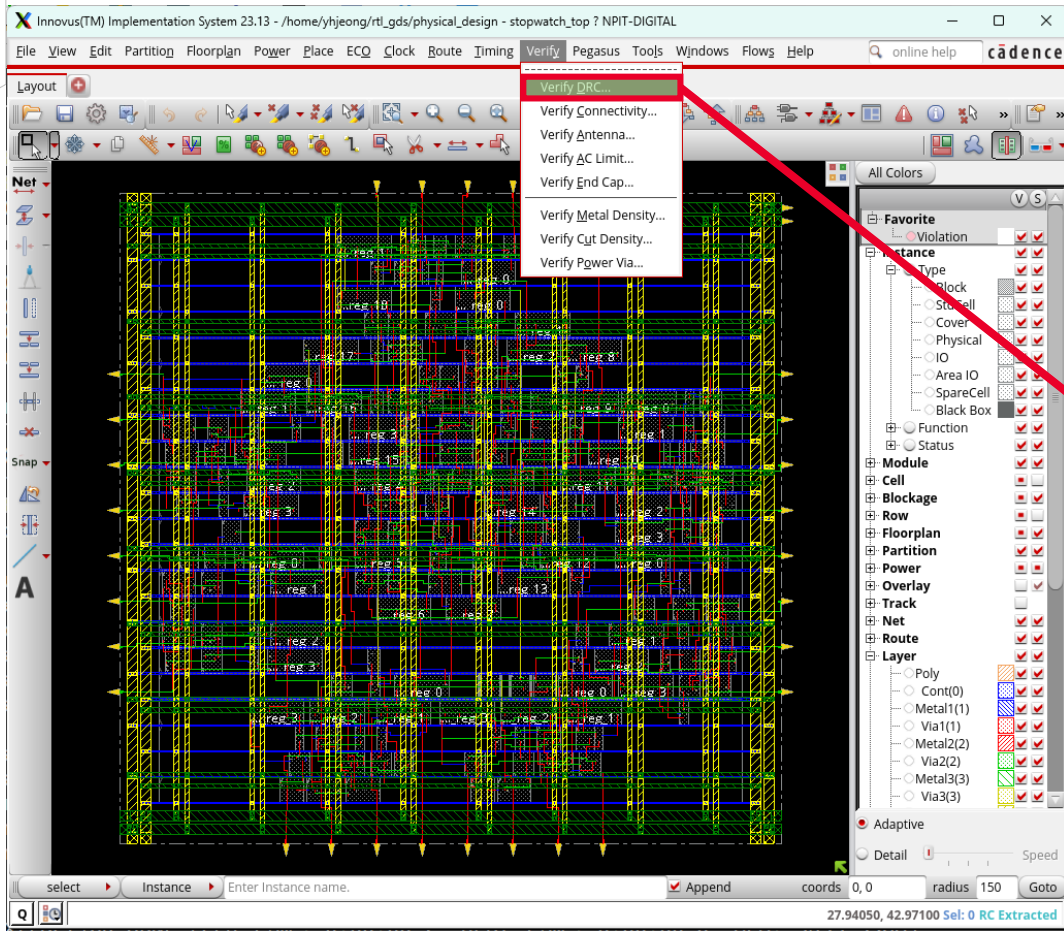
Route

Nano Route

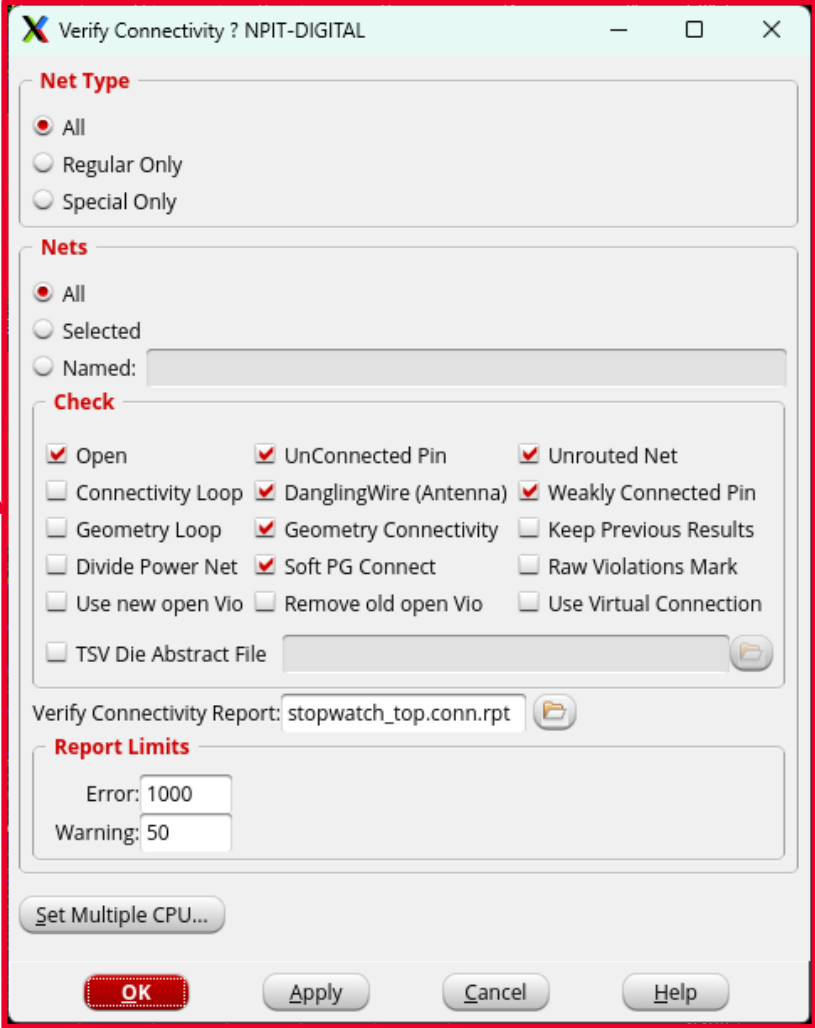
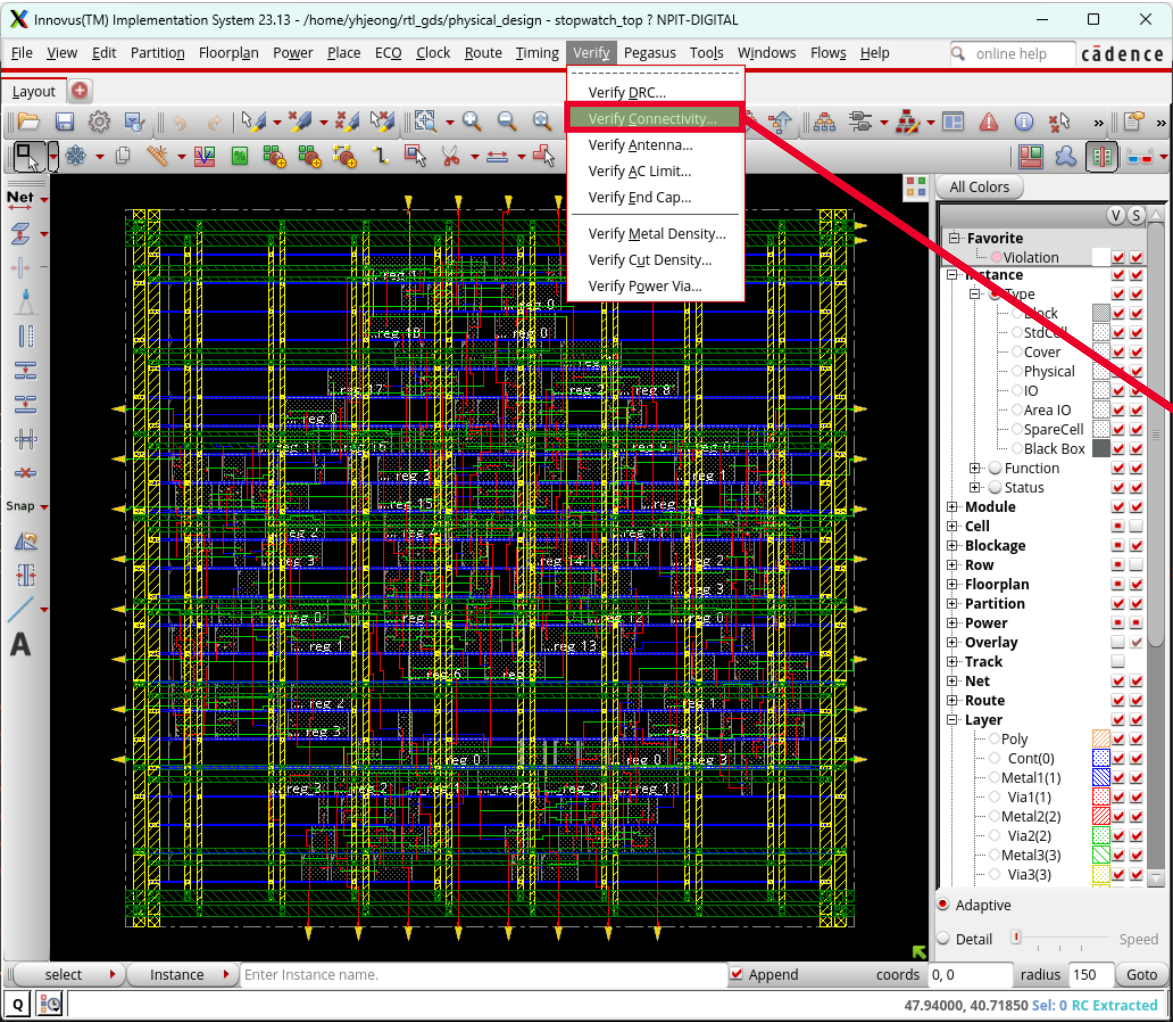


Extract RC





Connectivity



time_Design -postRoute

timeDesign Summary			
Setup views included: FUNC_SS_WORST			
Setup mode	all	reg2reg	default
WNS (ns):	4.939	4.939	6.839
TNS (ns):	0.000	0.000	0.000
Violating Paths:	0	0	0
All Paths:	114	45	95
DRVs	Real		Total
	Nr nets(terms)	Worst Vio	Nr nets(terms)
max_cap	0 (0)	0.000	0 (0)
max_tran	0 (0)	0.000	0 (0)
max_fanout	0 (0)	0	0 (0)
max_length	0 (0)	0	0 (0)
Density: 45.219%			

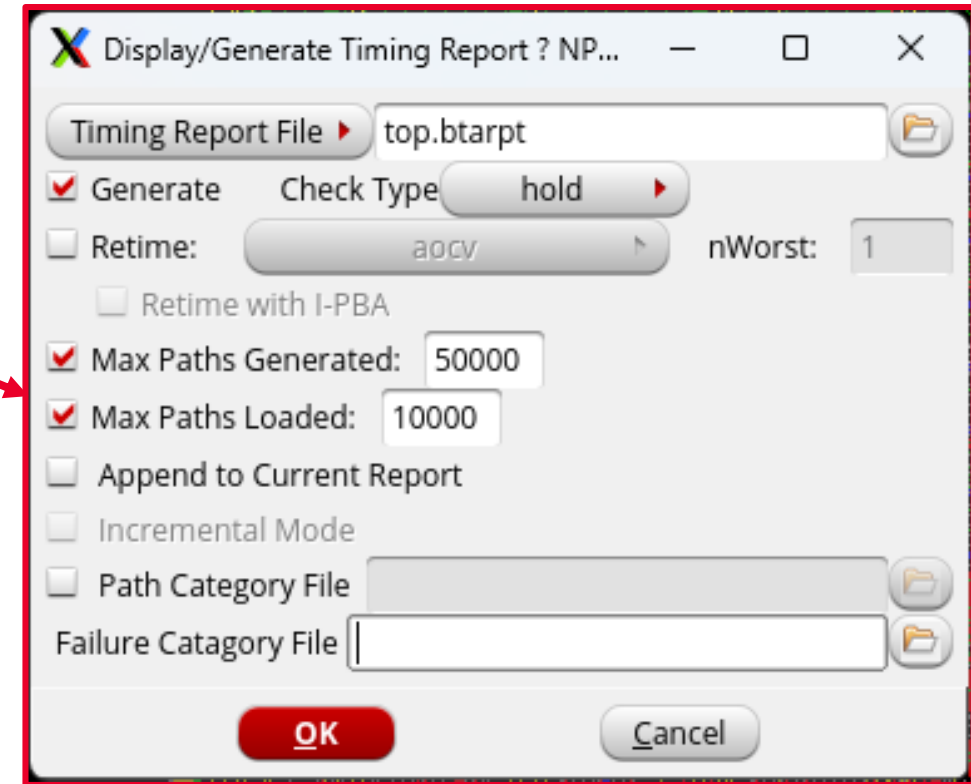
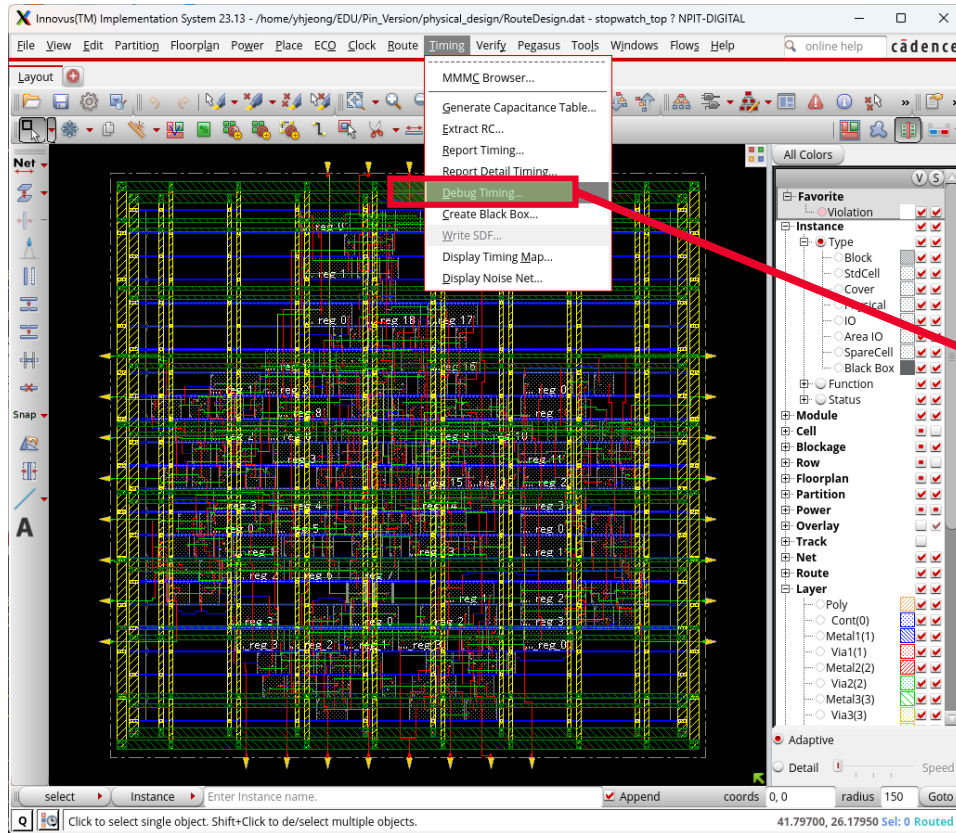
선택 후 실행

setDelayCalMode -SIAware false

time_Design -postRoute -hold

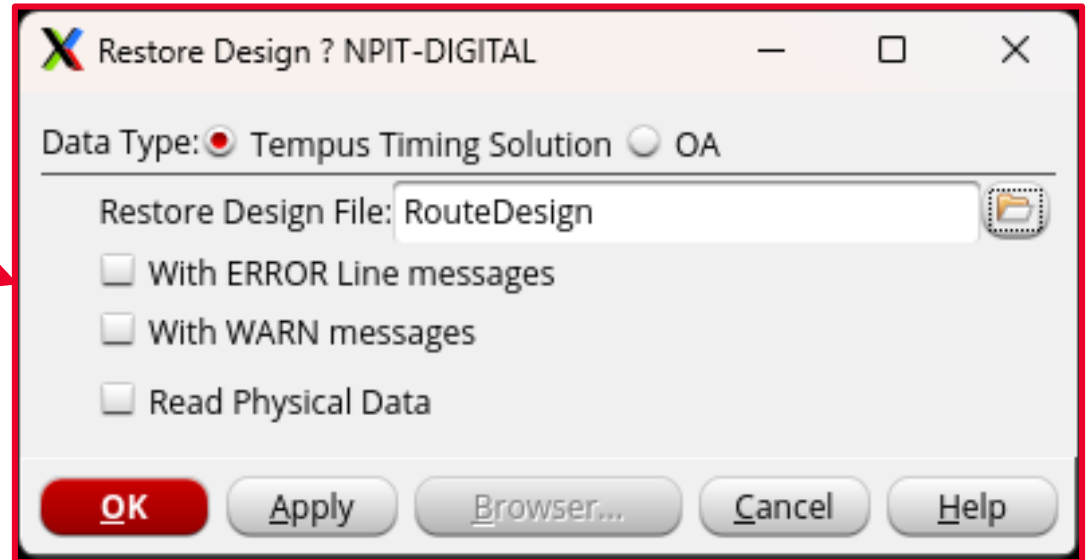
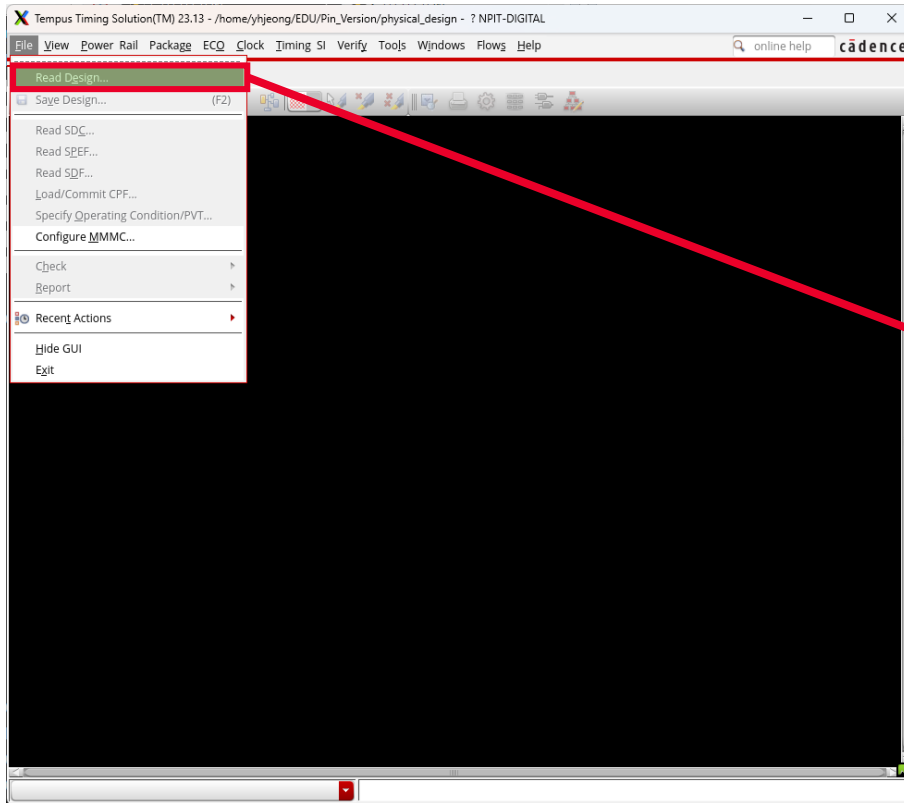
timeDesign Summary				
Hold views included: FUNC_FF_BEST				
Hold mode	all	reg2reg	default	
WNS (ns):	-0.030	0.084	-0.030	
TNS (ns):	-1.172	0.000	-1.172	
Violating Paths:	45	0	45	
All Paths:	114	45	95	
Density: 45.219%				

Debug Timing



saveDesign -rc -def RouteDesign

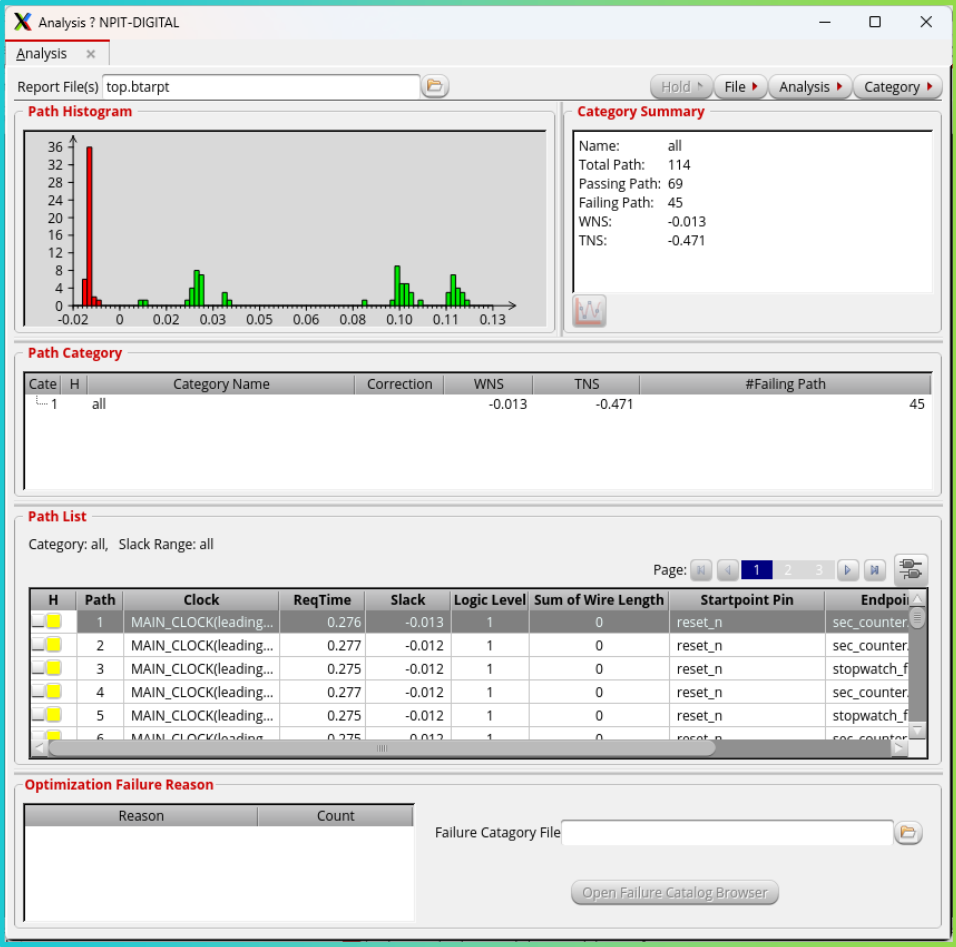
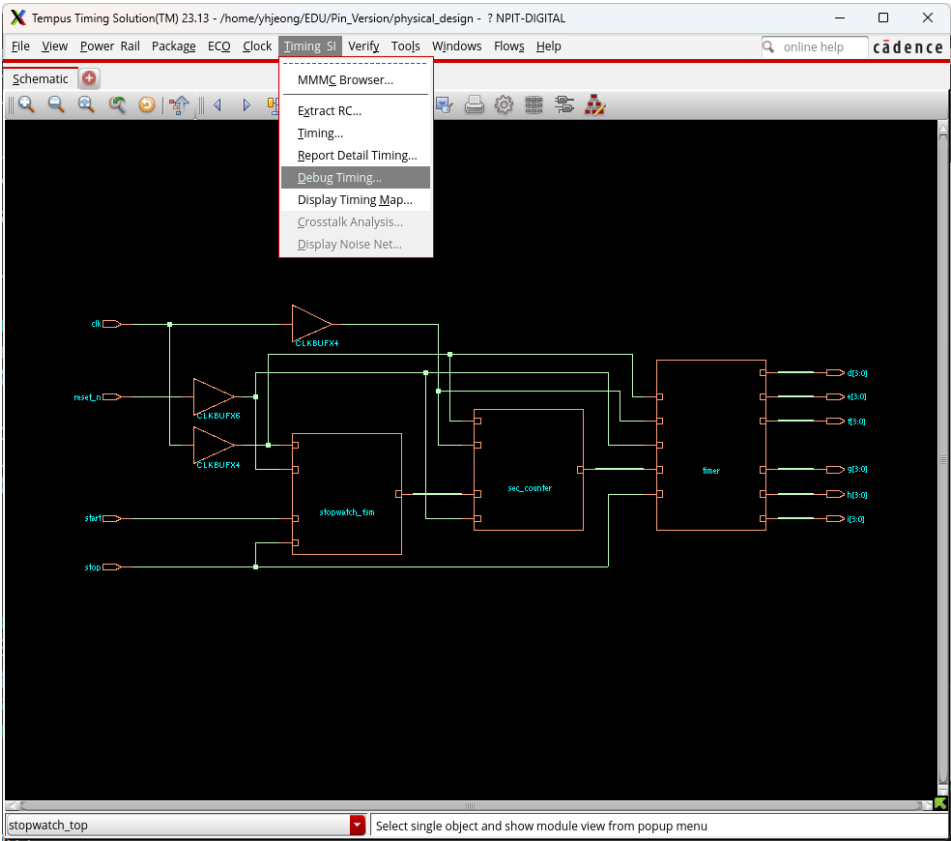
Read Design



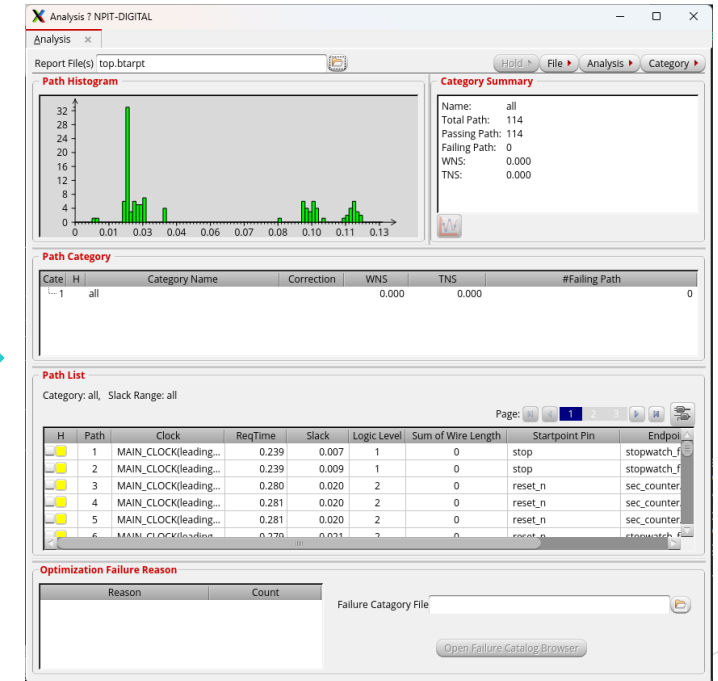
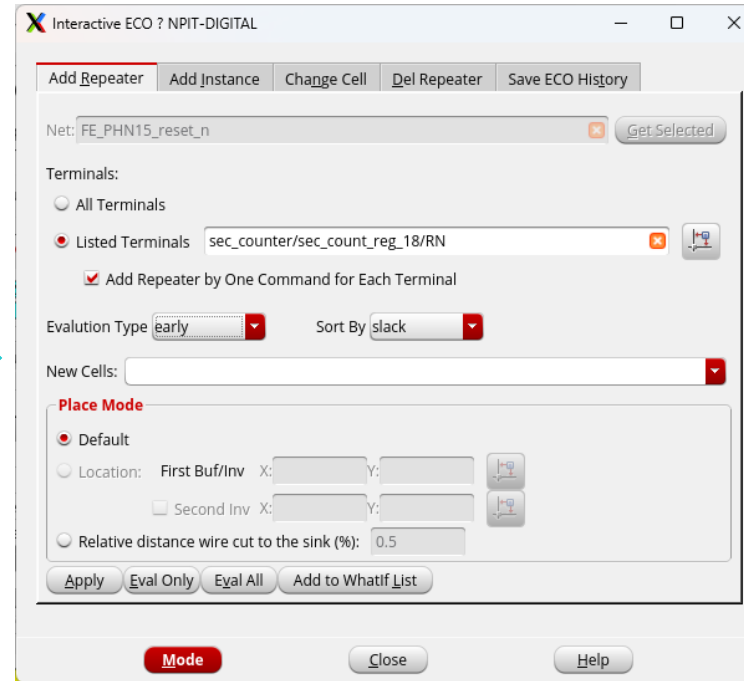
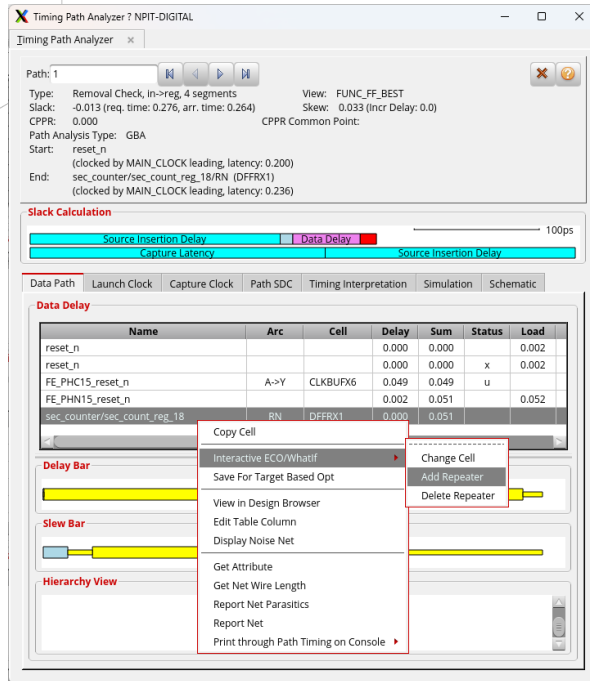
순서대로 입력

```
check_timing  
report_annotated_parasitics  
report_analysis_coverage  
report_analysis_coverage -check_type data_setup -verbose untested -exclude_untested const
```

Analysis View



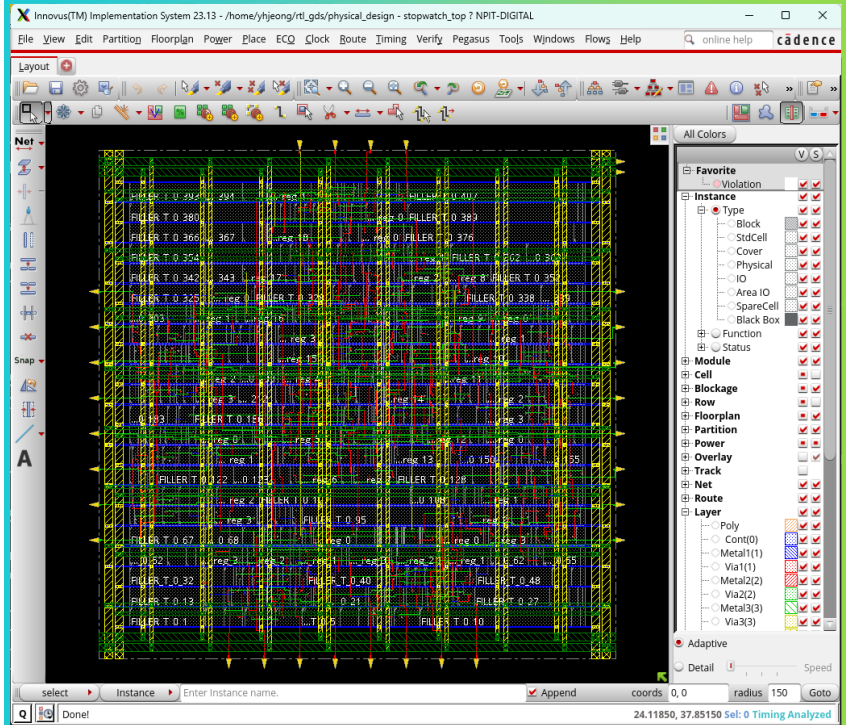
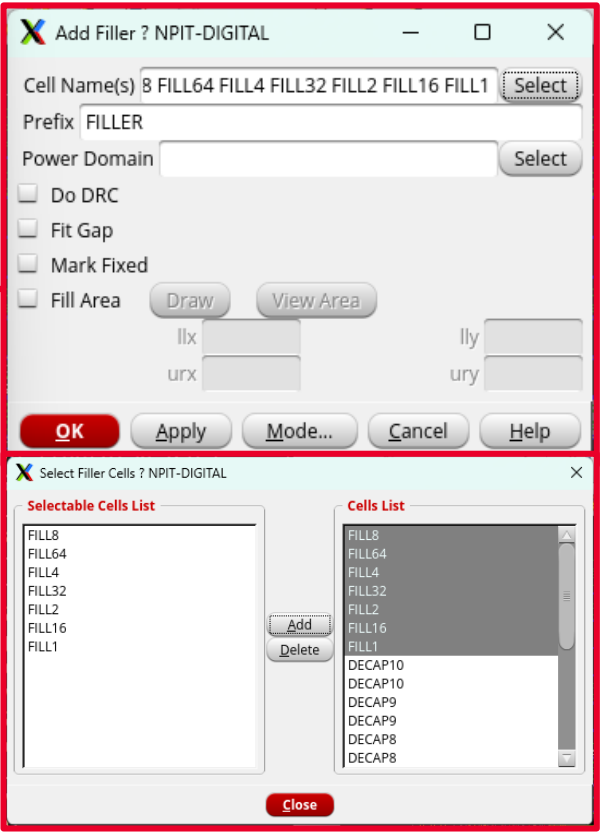
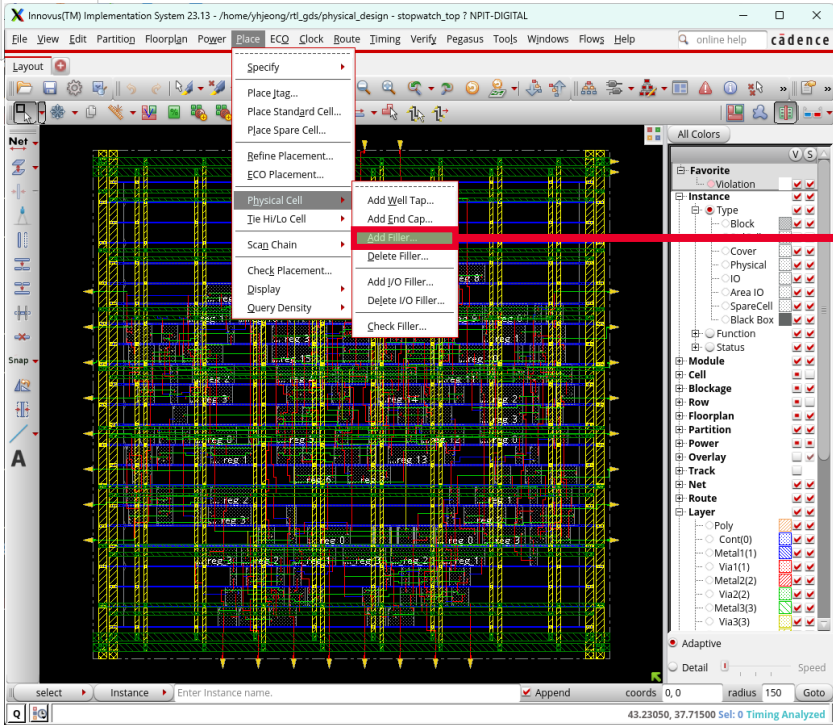
Add Repeater



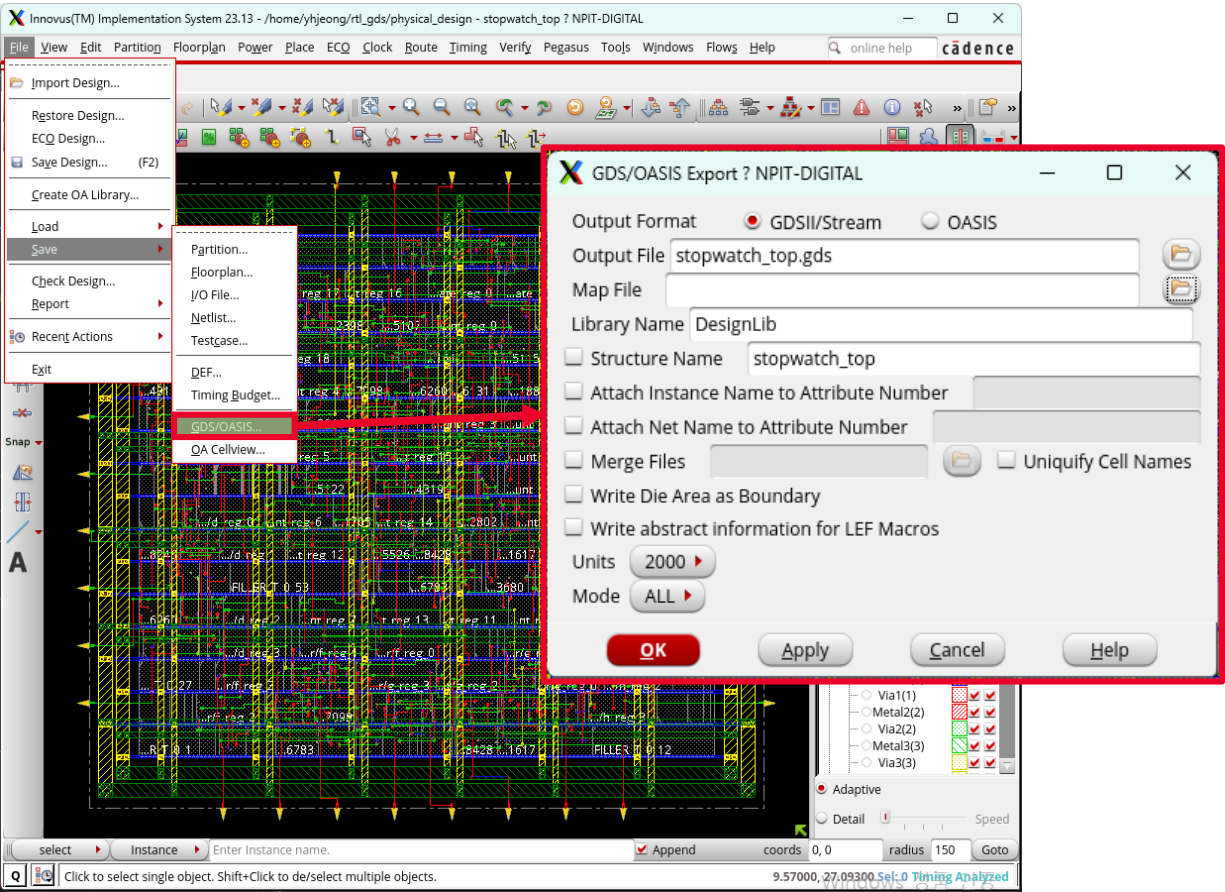
순서대로 입력

report_analysis_coverage
write_eco -format innovus -output innovus.eco

Chip Finish



GDS Export



```
Special Nets      83
  metal layer Metal1    51
  metal layer Metal10   16
  metal layer Metal11   16

Via Instances     1352

Metal Fills       0
  Via Instances   0
Metal FillOPCs    0
  Via Instances   0
Metal FillDRCs    0
  Via Instances   0
Text              273
  metal layer Metal1    4
  metal layer Metal2   183
  metal layer Metal3    83
  metal layer Metal4    1
  metal layer Metal10    2

Blockages         0

Custom Text       0

Custom Box        0

Trim Metal        0

#####Streamout is finished!
```